



جامعة السلطان قابوس **Sultan Qaboos University**

Analysis and Forecasting Oman Oil Prices Using Statistics Models and machine learning techniques

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Abstract

This study aims to analyze and forecast oil prices in Oman using both statistical and machine learning methods. The dataset covers the period from 2016 to 2022 and includes oil prices, exports, and production. The analysis begins with descriptive statistics and exploratory data analysis to better understand the behavior of the variables. A multiple linear regression model is used to examine the relationship between oil prices and the explanatory variables. The results show that production has a significant positive effect on oil prices, while exports do not show consistent significance.

Time series analysis is then applied using the Box–Jenkins methodology. The results indicate that the oil price series is non-stationary, but it becomes stationary after applying first differencing. Based on the ACF and PACF analysis, the ARIMA(0,1,1) model is selected and estimated. The model provides reasonable forecasts, although the accuracy is moderate.

In addition, a Support Vector Regression (SVR) model is applied; however, it produces relatively high prediction errors when used alone. To improve forecasting performance, a hybrid SVR-ARIMA model is developed. The results show that the hybrid model performs better than both ARIMA and SVR models, as it achieves lower RMSE values and follows the actual oil price movements more closely.

Overall, the findings suggest that combining statistical models with machine learning techniques can improve forecasting accuracy. The hybrid SVR-ARIMA model is therefore recommended as the most suitable approach for oil price prediction in this study, as it provides useful insights for economic planning and decision-making in Oman.

Chapter 1

Introduction

Introduction

Oil is a vital natural resource worldwide and plays a key role in the economic development of many countries. In Oman, oil is the main source of income and contributes significantly to government revenue. Therefore, fluctuations in oil prices can have a major impact on the Omani economy, affecting budgeting, investment, and long-term planning.

Oil prices are highly volatile and can change due to several factors, including global demand, supply conditions, political events, economic changes, and decisions made by oil-producing countries. For example, economic crises or changes in production levels can lead to sharp increases or decreases in oil prices. This makes it difficult for governments and businesses to accurately predict future oil prices.

To address this challenge, various forecasting methods have been developed. Traditional statistical techniques, especially time series models such as ARIMA and Exponential Smoothing, are widely used to identify trends and patterns in oil prices, which help in making predictions.

In recent years, machine learning methods such as regression models, Random Forest, and Support Vector Regression have gained attention for improving prediction accuracy. These methods are capable of handling complex relationships and capturing non-linear patterns that traditional models may not detect.

This study aims to forecast oil prices in Oman by comparing time series and machine learning methods to determine which approach provides more accurate predictions. The goal is to provide useful insights into oil price movements and support better decision-making for economic planning in Oman.

The Importance of Oil in Oman's Economy

Oil plays a central role in the economy of Oman and is considered the backbone of the country's economic growth. Since the discovery of oil in the 1960s, the oil sector has been the primary source of national income and government revenue. It contributes significantly to Oman's Gross Domestic Product (GDP), exports, and overall public finances.

One of the most important roles of oil in Oman's economy is its contribution to government revenue. A large portion of the government's budget is generated from oil exports. This revenue is used to support public services such as education, healthcare, infrastructure development, and social welfare programs. Therefore, fluctuations in oil prices have a direct impact on government spending and economic stability.

Oil exports also represent a major share of Oman's total exports. The country relies heavily on exporting crude oil to international markets, making it an important participant in the global energy market. When oil prices increase, export revenues rise, while lower prices reduce national income and may lead to economic challenges.

In addition, the oil sector contributes to economic growth by creating employment opportunities and encouraging investment. It has supported the development of related industries such as transportation, construction, and energy services. Furthermore, oil revenues have been invested in diversification projects aimed at reducing Oman's dependence on oil in the future.

However, this heavy reliance on oil makes the economy vulnerable to global price fluctuations. A sharp decline in oil prices can lead to budget deficits and economic uncertainty. For this reason, understanding and forecasting oil price movements is essential for effective economic planning and policy-making.

In conclusion, oil remains a vital component of Oman's economy, influencing its growth, stability, and development. A clear understanding of oil price trends is necessary to support sustainable economic progress.

oil company in Oman

Oman has several key oil companies, with Oman Oil Company (OOC) being one of the most prominent. Established in 1996, OOC plays an important role in managing the country's oil resources, with a focus on exploration, production, and distribution of petroleum products. It is wholly owned by the government and contributes significantly to the national economy.

In addition, Petroleum Development Oman (PDO), a joint venture between the government and international oil companies, is the largest oil producer in the country. These companies are central to Oman's oil industry, supporting economic growth, creating employment opportunities, and generating national revenue through oil exports.

Challenges Faced by the Oil Sector in Oman

The oil sector in Oman faces several challenges that affect its growth and stability. One of the main challenges is the fluctuation of global oil prices, which can lead to significant changes in revenue. Since Oman depends heavily on oil exports, any decline in prices can result in budget deficits and economic instability. Another important challenge is the limitation of oil reserves, as production is expected to decline in the coming years, raising concerns about long-term sustainability.

In addition, technological limitations in oil exploration and extraction can reduce efficiency and productivity. The sector also faces environmental challenges, including the need to comply with regulations and adopt cleaner technologies to reduce carbon emissions. Furthermore, geopolitical factors, such as tensions in the Middle East, can disrupt oil trade and supply chains, which may negatively affect the performance of the oil industry.

Research Problem

The oil sector in Oman plays a vital role in the country's economy, contributing significantly to government revenue and overall economic development. However, the industry faces several challenges, including fluctuations in oil prices, limited reserves, and environmental concerns. Accurate forecasting of oil prices is essential for effective economic planning, yet traditional models may not fully capture the complex dynamics of the market. Therefore, this research aims to address the need for more accurate and reliable forecasting models for oil prices in Oman by comparing both statistical and machine learning approaches.

Research Questions

1. What are the key factors that influence oil price fluctuations in Oman?
2. How accurate are traditional statistical methods, like ARIMA, in forecasting Oman's oil prices?
3. Can machine learning techniques, such as regression and support vector machines, improve the accuracy of oil price forecasts in Oman?
4. Which forecasting method, statistical or machine learning, provides the most reliable predictions for Oman's oil prices?

Main Objective

The main objective of this study is to compare the effectiveness of traditional time series models and machine learning techniques in forecasting oil prices in Oman, and to identify the best approach for improving forecast accuracy.

Specific Objectives

1. To analyze historical data on Oman's oil prices, production, and exports.
2. To apply and evaluate traditional time series models, such as ARIMA, for forecasting oil prices in Oman.
3. To apply machine learning techniques, including regression models and support vector regression (SVR), for oil price prediction.
4. To compare the performance of statistical and machine learning models in terms of forecast accuracy.

Database

	years	month	prices	exports	productions	USD_to_OMR	date
1	2016	January	42.280	27327.52	31198.66	0.384299	2016-01-01
2	2016	February	34.590	25647.13	29397.68	0.385221	2016-02-01
3	2016	March	27.400	30968.93	30263.83	0.384871	2016-03-01
4	2016	April	30.320	26989.12	29829.10	0.384658	2016-04-01
5	2016	May	36.340	26939.21	30994.90	0.383950	2016-05-01
6	2016	June	39.400	26698.22	30369.64	0.383950	2016-06-01
7	2016	July	44.330	26145.40	31344.89	0.383793	2016-07-01
8	2016	August	46.600	25745.20	31392.84	0.385086	2016-08-01
9	2016	September	43.400	24952.29	30100.74	0.384662	2016-09-01
10	2016	October	44.020	27264.29	31368.65	0.384833	2016-10-01
11	2016	November	43.920	27771.65	30448.25	0.383733	2016-11-01
12	2016	December	49.180	25491.70	30850.92	0.385252	2016-12-01
13	2017	January	44.540	26650.14	26650.14	0.385032	2017-01-01
14	2017	February	52.720	22530.77	27252.25	0.384040	2017-02-01
15	2017	March	53.930	25573.69	29992.21	0.383991	2017-03-01
16	2017	April	55.120	22745.33	29017.26	0.383993	2017-04-01
17	2017	May	51.710	24813.94	30095.07	0.384187	2017-05-01
18	2017	June	52.820	24144.24	29416.97	0.384540	2017-06-01
19	2017	July	50.550	25440.81	29977.00	0.384391	2017-07-01
20	2017	August	46.520	25186.64	30011.38	0.384166	2017-08-01

```
tibble [84 x 7] (S3: tbl_df/tbl/data.frame)
 $ years      : int [1:84] 2016 2016 2016 2016 2016 2016 2016 2016 2016 2016 ...
 $ month      : Ord.factor w/ 12 levels "January"<"February"<...: 1 2 3 4 5 6 7 8 9 10 ...
 $ prices     : num [1:84] 42.3 34.6 27.4 30.3 36.3 ...
 $ exports    : num [1:84] 27328 25647 30969 26989 26939 ...
 $ productions: num [1:84] 31199 29398 30264 29829 30995 ...
 $ USD_to_OMR : num [1:84] 0.384 0.385 0.385 0.385 0.384 ...
 $ date       : Date[1:84], format: "2016-01-01" "2016-02-01" ...
```

- ☐ **Oil Prices:** USD per barrel
- ☐ **Exports:** Millions of USD
- ☐ **Production:** Barrels per day (bpd)

Literature review

1- COVID-19 Deaths Cases Impact on Oil Prices: Probable Scenarios on Saudi Arabia Economy.

This essay addresses how COVID-19 death cases have affected crude oil prices worldwide, emphasizing the increased volatility and investor anxiety. The study examines daily data from January 22 to June 14, 2020, using the ARDL econometric model. Saudi Arabia is most impacted by the death ratio's large negative impact on the dynamics of oil prices. The study highlights how risks associated with mortality are magnified for the actual economy and financial market.

2- Oil Price Volatility and Economic Growth Numerous studies have investigated the effects of oil price volatility on economic performance. For instance, Smith (2018) found that price spikes tend to increase production costs in oil-importing countries, leading to inflationary pressures. Conversely, Brown and Jones (2020) demonstrated that falling prices negatively affect public spending in oil-exporting nations, causing fiscal deficits.

3- On the relation between the crude oil market and pandemic Covid-19

The authors analyze the Covid-19 pandemic's overreaction on the crude oil market, revealing increased sensitivity to fake news and a high level of energy trader sentiment. They also show a close connection between the market and other asset classes, with the total connectedness index contributing 35% to spillover. The authors note that infectious diseases increase investor panic and anxiety. The author used T-test, logistic regression and time series to analyze the data.

4- A crude future? COVID-19 challenges oil demand, supply and prices.

This study shows how oil prices are expected to recover by Q3 2020 and modestly in Q1 2021, with WTI and Brent prices ranging \$40-560 per barrel. However, many oil sectors have faced losses due to production equipment access, costs, and market competitiveness.

5- COVID-19 and global oil markets

This study says that due to the pandemic and the "Oil Price War" between Saudi Arabia and Russia in March 2020, the price of oil dropped globally, forcing producers to find storage space. The West Texas Intermediate price entered negative territory, underscoring the interrelated factors influencing the structure and future power of the global oil industry.

6- The impact of the war between Russia and Ukraine on oil exports

Russia is one of the largest oil exporters globally, contributing approximately 10–12% of the world's supply:

BP Statistical Review (2021): Russia accounted for over 7 million barrels per day (b/d) of crude oil and refined products exports, with Europe being the primary destination. And IEA (2022): Prior to the war, Russian crude accounted for 27% of Europe's imports, making the region heavily reliant on Russian supplies.

Time-Series Analysis To examine trends in oil exports and price volatility before and after the war. The literature emphasizes the critical role of Russian oil in stabilizing global markets, with disruptions creating significant ripple effects across economies.

7-Impact of Global Oil Prices on Oman's Economy

The volatile nature of global oil prices between 2016 and 2022 posed significant challenges to Oman's economy. Scholarly works have addressed:

Economic Vulnerability: Studies highlight Oman's reliance on oil revenues for its fiscal budget, noting that price shocks lead to budget deficits and economic adjustments, including subsidy reforms and fiscal consolidation measures (IMF, 2021).

Policy Responses: Research underscores Oman's diversification initiatives under Vision 2040, aiming to reduce dependency on oil while promoting other economic sectors like tourism, logistics, and manufacturing (Al-Zakwani, 2021).

Data Cleaning and Preparation

1- Create a proper monthly date column

The dataset is transformed by combining the separate year and month columns into a single date variable. This new date column represents the first day of each month and allows the data to be properly structured for time series analysis and visualization. In addition, the dataset is arranged in chronological order to ensure accurate analysis and interpretation.

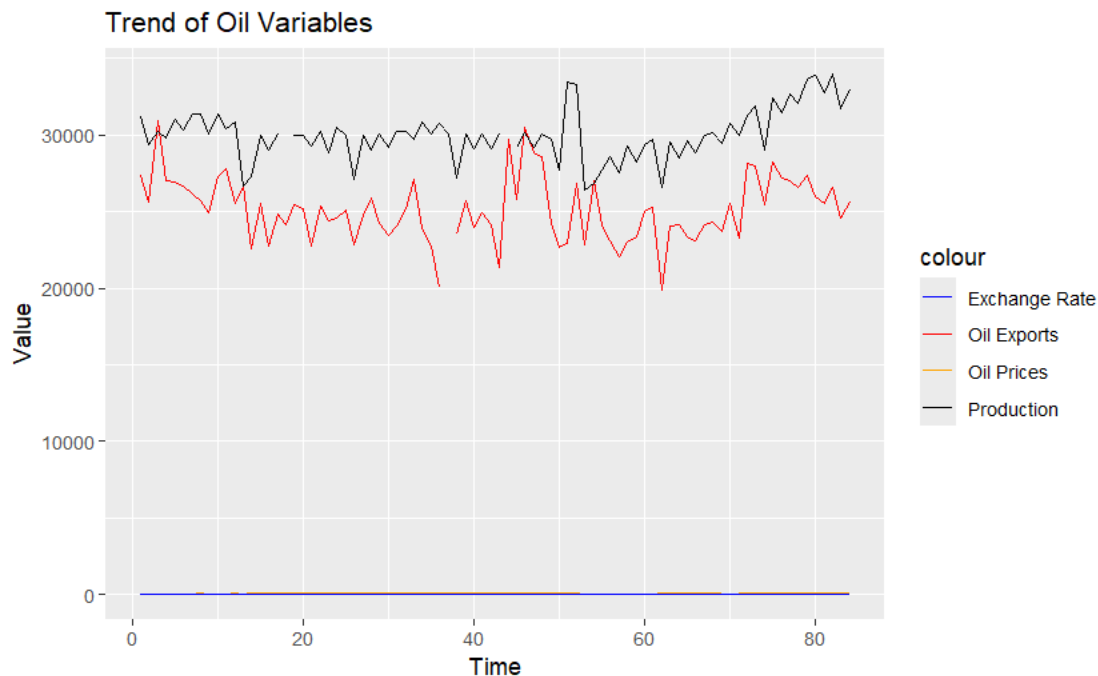
2- Check missing values

Missing values before imputation:

years	month	prices	exports	productions	USD_to_OMR	date
0	0	1	1	2	0	0

years <int>	month <ord>	prices <dbl>	exports <dbl>	productions <dbl>	USD_to_OMR <dbl>	date <date>
2017	June	52.82	24144.24	NA	0.384540	2017-06-01
2019	January	66.30	NA	30078.30	0.384187	2019-01-01
2019	August	61.70	29741.60	NA	0.385155	2019-08-01
2021	October	NA	25524.09	30774.38	0.385279	2021-10-01

4 rows



It is important to note that some values are missing, particularly in the exports, production, and prices columns for certain months. For example, there is no oil price data for October 2021, and no export data for January 2019. These missing values require appropriate handling, such as

imputation using monthly averages or other suitable methods. The newly created date column allows for easier chronological analysis of the data.

The imputation of missing values based on the average of each month across all years provides a more complete dataset. This is essential for improving the accuracy of forecasts and ensuring reliable time series analysis.

3- Impute missing values based on the monthly average from all years

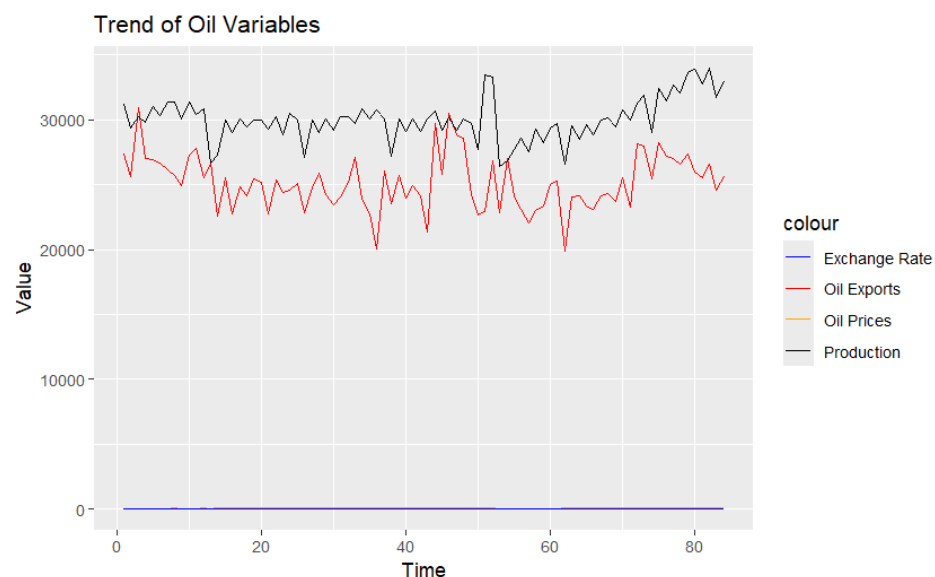
The dataset is processed to handle missing values by imputing them based on the monthly average across all years. This method ensures that each missing value is replaced with the average value for the corresponding month, calculated from the available data. This approach helps maintain the consistency of the dataset and preserves the seasonal patterns in the monthly data.

After performing the imputation, the results are checked by displaying the first few rows of the modified dataset and confirming that no missing values remain. This step ensures that the imputation process is successful and that the dataset is complete.

Subsequently, the trends of key variables, including oil prices, exports, production, and exchange rates, are visualized using line plots. These plots reveal patterns and relationships between the variables and provide insights into how each variable changes over time.

This process of imputing missing values and visualizing the data is essential for maintaining data integrity and for developing a clearer understanding of the behavior of oil prices and related variables over time.

```
Missing values after imputation:
years      month      prices      exports      productions      USD_to_OMR      date
0          0          0          0          0          0          0
```



EDA

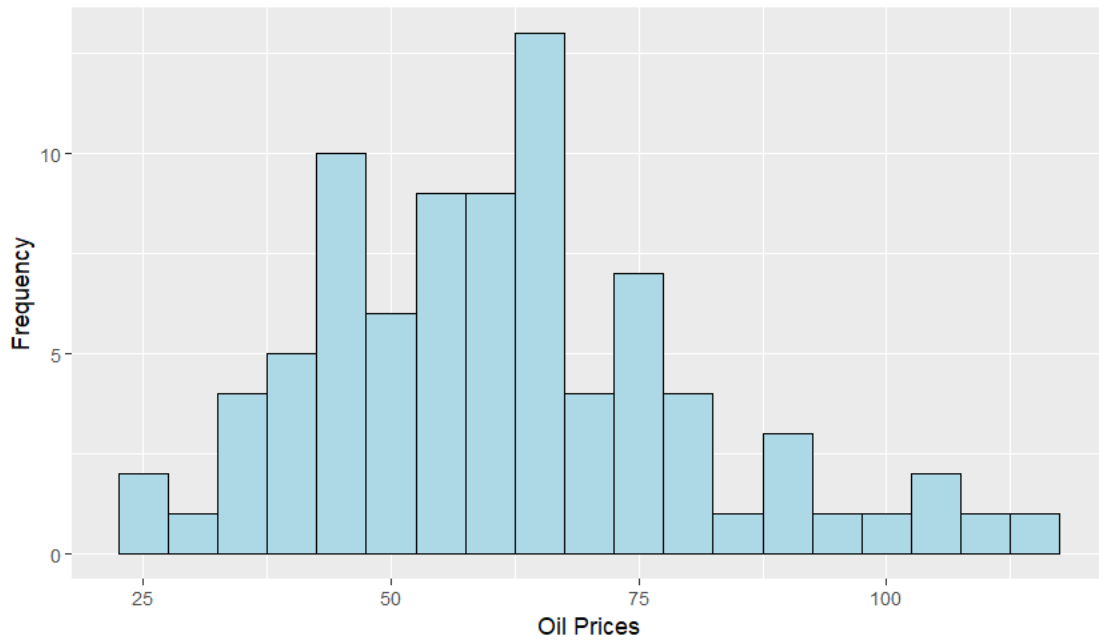
Data Structure:

- The dataset consists of 84 rows and 7 columns.
- The columns include:
 - years: A numerical column indicating the year.
 - month: An ordered factor column, where months are ordered from January to December.
 - prices: Numerical values representing oil prices.
 - exports: Numerical values representing exports.
 - productions: Numerical values representing production.
 - USD_to_OMR: Numerical values representing the exchange rate between USD and OMR.
 - date: A Date column, formatted as "YYYY-MM-DD."

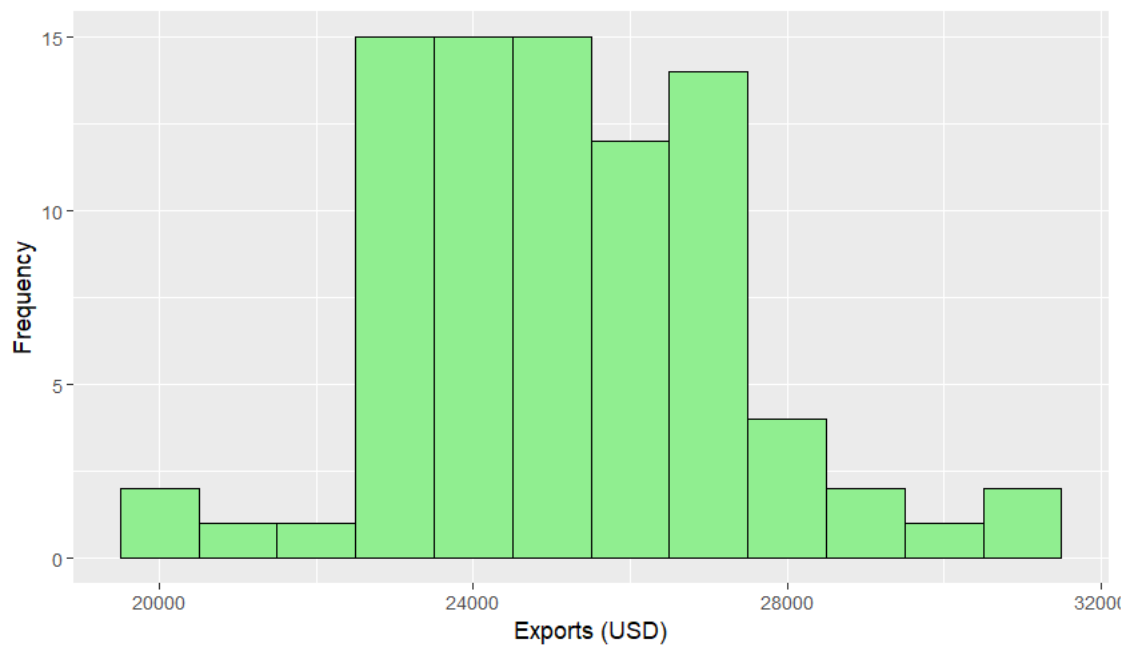
Statistical Summary:

- years: The years in the dataset range from 2016 to 2022.
- month: The months in the dataset are well-distributed across the 12 months of the year.
- prices:
 - The minimum value is 23.65 (in 2016), and the maximum value is 112.93 (in 2022).
 - The median value is 61.10, and the mean is 59.99.
- exports:
 - The minimum export value is 19870 in 2016, and the maximum is 30969 in 2022.
 - The median is 25126, and the mean is 25149.
- productions:
 - The production values range from 26374 (minimum) to 33944 (maximum).
 - The median and mean production values are 30023 and 30022, respectively.
- USD_to_OMR:
 - The exchange rate ranges from 0.3837 (minimum) to 0.3853 (maximum), with a mean of 0.3844.

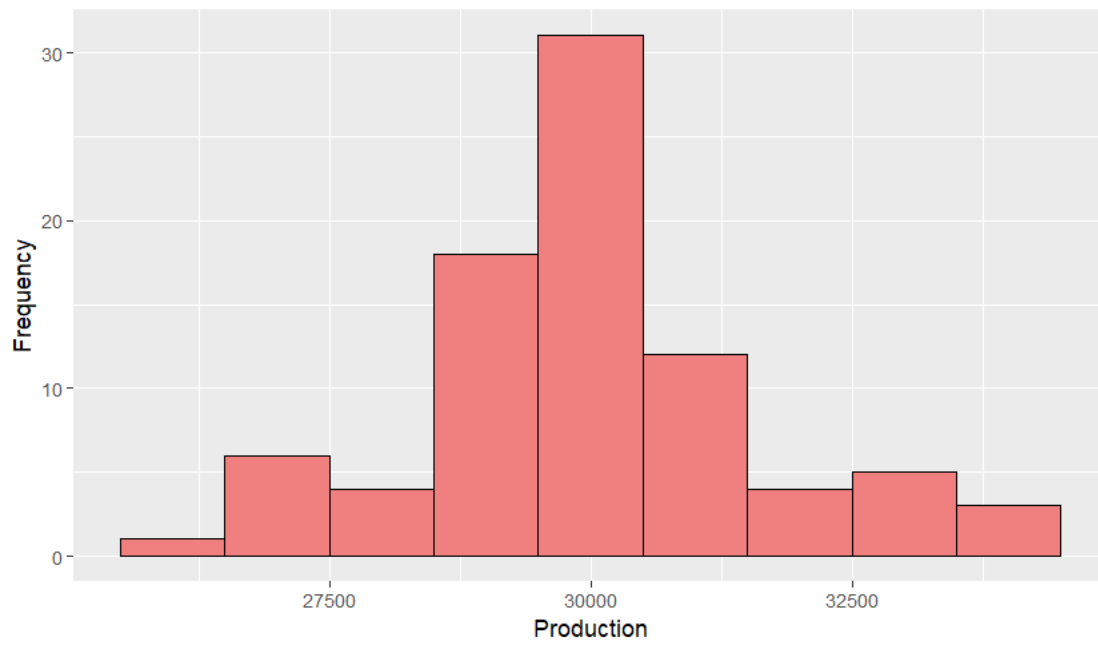
distribution of oil prices

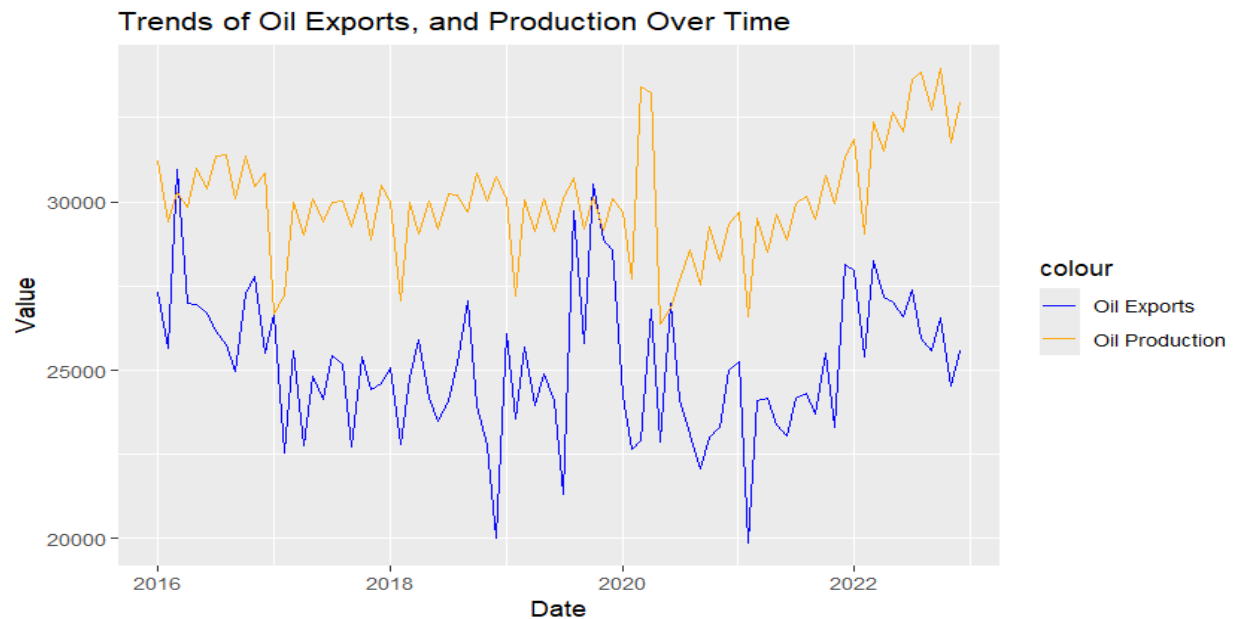


distribution of exports



distribution of production





Oil Exports (Blue Line):

- The oil export values fluctuate over time, with noticeable peaks and troughs.
- There is a general upward trend starting from around 2020, indicating that exports have increased in the later years.
- The fluctuations show seasonal or cyclical changes, possibly influenced by global demand or domestic factors such as production capacity.

Oil Production (Orange Line):

- Similarly, oil production shows variations over time, with significant peaks around 2020 and gradual increases afterward.
- Production seems to have more consistent periods of increase after 2019, suggesting that the country may have been ramping up production during this time.

Correlation Between Exports and Production:

- The two lines appear to have a positive correlation, meaning that as oil production increases, oil exports also tend to rise. This could indicate that the country is producing more oil and successfully exporting the surplus.
- The slight divergence between the two variables, especially around certain peaks (like in 2020), might suggest external factors such as market demand or logistical constraints.

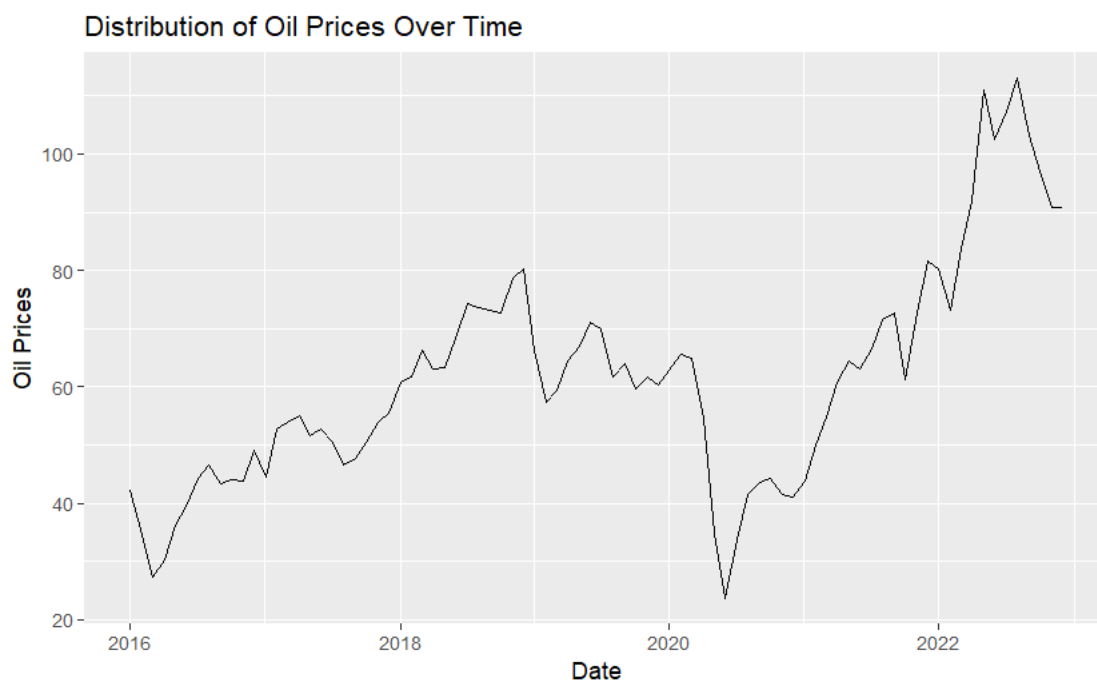
The plot provides valuable insights into the relationship between oil production and oil exports over time. It highlights periods where both production and exports increase, particularly in recent years, possibly due to changes in market conditions or production strategies. The upward trend observed from 2020 onwards indicates growth in the oil sector, with both production and exports increasing over time.

Oil Price Fluctuations:

- The plot indicates significant fluctuations in oil prices from 2016 to 2022.
- Prices began around 2016 at a relatively moderate value, around \$40 per barrel.
- A significant drop is observed in early 2020, likely due to the COVID-19 pandemic, which led to a global decrease in demand and an overall oil price slump.

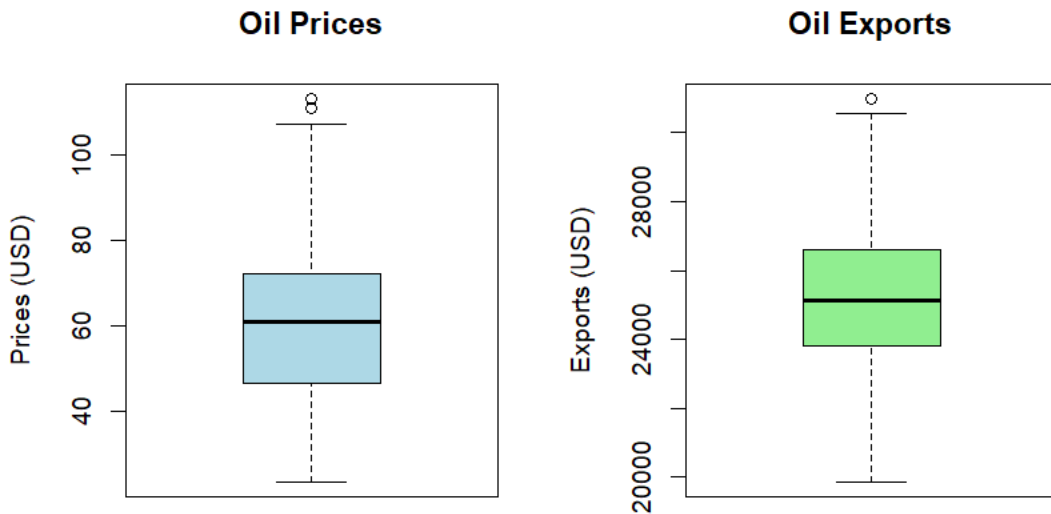
Recovery After 2020:

- After the sharp decline in 2020, oil prices started to increase steadily from 2021.
- The prices seem to peak in 2022, surpassing \$100 per barrel, suggesting a strong recovery and possibly reflecting the impact of geopolitical events or shifts in global demand.



This plot effectively shows how oil prices have evolved over time, highlighting key events that caused significant fluctuations, such as the pandemic-induced dip and the later recovery. The steady rise after 2020 suggests a positive outlook for the oil market, influenced by global demand dynamics, economic recovery, and geopolitical factors.

box plots

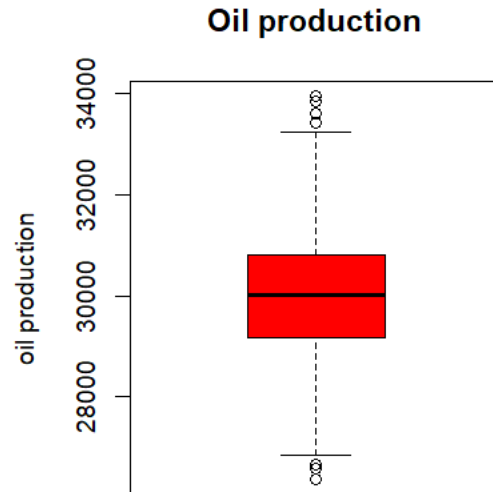


Oil Prices - USD per Barrel:

- The oil prices range from approximately \$40 to \$110 per barrel.
- Interquartile Range (IQR): The middle 50% of the data lies between the first quartile (around \$50) and the third quartile (around \$85), indicating a typical price range.
- Median: The median oil price is around \$60, which is the central tendency of the prices over the given period.
- Outliers: The plot shows a few outliers (circles above \$100), which represent unusually high oil prices, possibly during periods of significant market disruption or geopolitical factors affecting oil prices.

Oil Exports - Millions of USD:

- The oil exports range from approximately \$20,000 to \$30,000 million (or \$20–30 billion).
- Interquartile Range (IQR): The middle 50% of the export values lies between the first quartile (around \$24 billion) and the third quartile (around \$28 billion).
- Median: The median export value is around \$25 billion, indicating a central value for exports over time.
- Outliers: There are outliers above \$28 billion in the export data, indicating exceptionally high export values during specific periods.



Range of Production:

- The oil production values range from approximately 28,000 to 34,000 barrels per day.
- The box shows the interquartile range (IQR), with the middle 50% of the data lying between 29,000 bpd and 31,500 bpd.

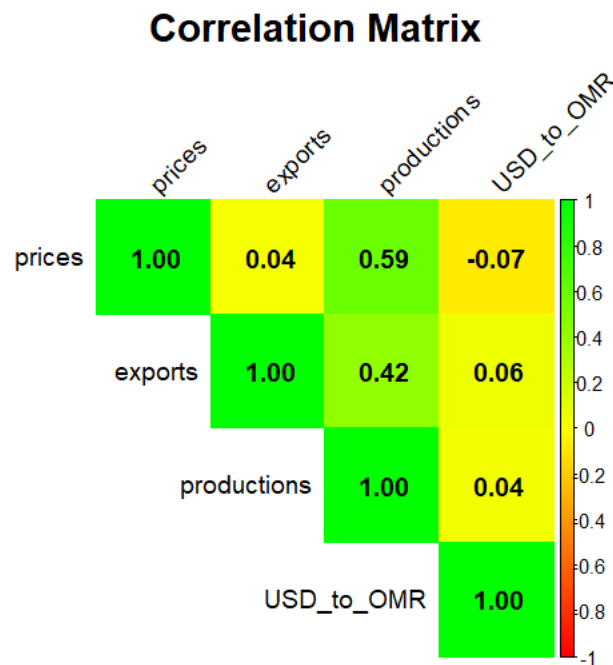
Median Production:

- The median value for oil production, shown as the central line inside the box, is around 30,000 bpd, indicating the typical daily production over the observed period.

Outliers:

- The plot indicates several outliers (circles above 34,000 bpd), which represent days with unusually high production values compared to the rest of the data. These outliers could indicate special events, such as temporary production boosts due to external factors like geopolitical events, technological advancements, or changes in production strategies

Correlation



Oil Prices (prices):

- Prices vs Exports: The correlation between oil prices and exports is very low (0.04), indicating that changes in oil prices have little to no correlation with the value of exports in USD. This suggests that oil prices may not directly influence export values in this dataset.
- Prices vs Production: The correlation between oil prices and production is moderate (0.59). This suggests that there is a positive relationship, where higher oil prices may be associated with higher production levels, likely due to increased demand or profitability.
- Prices vs USD to OMR: The correlation between oil prices and the USD to OMR exchange rate is slightly negative (-0.07), suggesting a very weak inverse relationship. This might indicate that when oil prices rise, the OMR might weaken slightly, but the correlation is too small to be significant.

Oil Exports (exports):

- Exports vs Production: There is a moderate positive correlation (0.42) between oil exports and production. This indicates that as production increases, exports tend to rise as well, which is logical since higher production can lead to higher exports.
- Exports vs USD to OMR: The correlation is very low (0.06), indicating that the export values in USD have almost no relationship with the USD to OMR exchange rate.

Oil Production (productions):

- Production vs USD to OMR: The correlation here is very low (0.04), indicating that oil production levels have little to no relationship with the USD to OMR exchange rate.

USD to OMR (exchange rate):

- USD to OMR shows no significant correlations with the other variables, as most correlations are very weak.

Chapter 2

Methodology

In this project, several statistical methods are applied to ensure the use of appropriate techniques for the analysis. A range of approaches is selected to achieve the objectives of the study. The analysis begins with descriptive statistics, followed by the application of different statistical tests to examine the data.

2.1 One-Way ANOVA and Kruskal-Walli's test:

The One-Way Analysis of Variance (ANOVA) is a parametric test used to examine whether the mean values of a continuous variable differ across multiple groups defined by a categorical variable (e.g., comparing oil prices across different years or months).

2.2 Time series analysis:

Time series analysis involves studying data points collected or recorded at specific time intervals. This analysis helps identify patterns, trends, and seasonal behaviors in the data, and is widely used in fields like economics, finance, and environmental studies. Time series models aim to forecast future values based on historical data.

2.2.1 Definition of Time Series

A time series is a sequence of data points ordered in time. Each data point represents an observation or measurement taken at a specific time or interval. Time series data can be continuous (e.g., temperature over time) or discrete (e.g., monthly sales figures). The primary goal of time series analysis is to extract meaningful insights about the data's underlying structure to make predictions.

Components of Time Series Data:

1. **Trend:** The long-term movement in the data, which could be increasing, decreasing, or stable.
2. **Seasonality:** Patterns that repeat at regular intervals, often within a year or over a specific period (e.g., sales spikes during holidays).
3. **Cyclic Patterns:** Long-term fluctuations that are not strictly periodic and are often tied to economic or business cycles.
4. **Irregular/Random Component:** The residual noise or variability in the data that cannot be explained by the trend, seasonality, or cycles.

2.2.2 Box-Jenkins Methodology

The Box-Jenkins methodology is a popular approach for modeling and forecasting time series data. This method involves the following steps:

1. Identification:

- Plotting the Data: First, visualize the time series data to understand its components (trend, seasonality, etc.).
- Stationarity Check: Time series data should be stationary, meaning that its statistical properties (mean, variance) do not change over time. If the data is not stationary, transformations (such as differencing) may be needed.

There are two patterns of the Box-Jenkins models:

- **Seasonal**

- **non-seasonal.**

	SAC (Sample Autocorrelation Function)	SPAC (sample partial autocorrelation)
Definition	Measures the correlation between observations at different lags.	Measures the correlation between observations at different lags, adjusted for intermediate lags.
Purpose	Identifies how past values influence current values directly and indirectly.	Identifies direct influence of lagged terms on current values.
Used to Identify	Primarily helps in identifying the MA (q) part of ARIMA models.	Primarily helps in identifying the AR (p) part of ARIMA models.
	- SAC at lag k denoted r_k: $r_k = \frac{\sum_{t=k}^n (z_t - \bar{z})(z_{t-k} - \bar{z})}{\sum_{t=k}^n (z_t - \bar{z})^2}$ Where $\bar{z} = \frac{\sum_{t=b}^n z_t}{n - b + 1}$ - The standard error of r_k is: $S_{r_k} = \left(\frac{1 + 2 \sum_{j=1}^{k-1} r_j^2}{n - b + 1} \right)^{1/2}$ - The test statistic is: $t_{r_k} = \frac{r_k}{S_{r_k}}$ When: • $t_{r_k} > 1.6$ for low lags (1, 2, 3) • $t_{r_k} > 2$ for large lags (4, 5, ..., k) → That means there is spike at lag k	- SPAC at lag k denoted by: $r_{kk} = \begin{cases} r_1, & k = 1 \\ \frac{r_k - \sum_{j=1}^{k-1} r_{k-j} r_{j-1,k-j}}{1 - \sum_{j=1}^{k-1} r_{k-j}^2}, & k = 2, 3, \dots \end{cases}$ $r_{kj} = r_{k-1,j} - r_{kk} r_{k-1,k-j}$ - The standard error of r_{kk} is: $S_{r_{kk}} = \left(\frac{1}{n - b + 1} \right)^{1/2}$ - The $t_{r_{kk}}$ statistic is: $t_{r_{kk}} = \frac{r_{kk}}{S_{r_{kk}}}$ → There is a spike when $t_{r_{kk}} > 2 \text{ at lags } (1, 2, \dots, k)$

The Box–Jenkins forecasting approach begins with model identification by analyzing the behavior of the time series using:

- **SAC (Sample Autocorrelation Function)**
- **SPAC (Sample Partial Autocorrelation Function)**

A time series is considered **stationary** when its statistical properties, such as the mean and variance, remain stable over time and do not show continuous changes.

To assess stationarity, the SAC is commonly used, as it measures the relationship between current observations and their past values.

After confirming that the time series is stationary, an appropriate model can be selected, such as:

- **Autoregressive (AR) model**
- ARIMA model, which is a combined model that integrates autoregressive and moving average components and is suitable for handling non-stationary data.

Non-seasonal model:

model	SAC	SPAC
Autoregressive of order p (AR) model	Dies down	Cuts off after lag p
Moving Average of order p (MA) model	Cuts off after lag q	Dies down
ARIMA model	Dies down Quickly Cuts off after lag q	Dies down Quickly Cuts off after lag q

Autoregressive of order p (AR) model

$$z_t = \delta + \phi_1 z_{t-1} + \phi_2 z_{t-2} + \dots + \phi_p z_{t-p} + a_t$$

Where:

- z_t : current value of the time series
- $\phi_1, \phi_2, \dots, \phi_p$: autoregressive coefficients
- δ : constant (intercept)
- a_t : white noise/error term

-Parameter Estimation

Estimate the parameters (p, d, q) using statistical methods like Maximum Likelihood Estimation (MLE) or Least Squares.

testing of the constant and parameters in non-seasonal.

$H_0: \delta = 0$ $H_1: \delta \neq 0$ Test statistic: $t = s^z / \sqrt{(n - b + 1)}$ Where: $s^z = [\sum_{t=\beta}^n (z_t - \bar{z})^2 / (n - b + 1 - 1)]^{1/2}$ So: If $t > 2$, then δ (constant) should be included in the model.	$H_0: \theta = 0$ $H_1: \theta \neq 0$ Test statistic: $t = \hat{\theta} / s_{\theta}$ $s = \sqrt{[\sum_{i=1}^n (y_i - \hat{y}_i)^2 / (n - n_p)]}$ where n_p is number of parameters, and $\hat{\theta}$ is the point estimate of θ . s is the standard error of $\hat{\theta}$. So: Reject H_0 if $ t > t_{n-n_p, \alpha/2}$ p-value $<< \alpha$
---	---

Model Diagnostic Checking

Model diagnostic checking involves evaluating how well the time series model fits the data by examining the residuals (errors). The main purpose is to ensure that the residuals behave like random noise (white noise).

For the model to be considered appropriate, the residuals should:

- Be random, showing no clear pattern,
- Follow an approximately normal distribution,
- Have no autocorrelation, meaning they are not related to their past values.

2.2.3 Forecast

Forecasting involves predicting future values based on past and present data. Once a model is chosen and fitted using the Box-Jenkins methodology, it can be used to make predictions. Common forecasting models include ARIMA, (Seasonal ARIMA), and Exponential Smoothing.

1. ARIMA Model for Forecasting:

- The ARIMA model is a powerful tool for forecasting time series data. It is based on three key components:
 - AR (AutoRegressive): Relates current values to previous values.
 - I (Integrated): Accounts for non-stationarity by differencing the series.

2. Steps in ARIMA Forecasting:

- **Step 1: Stationarity Test:** Use ADF test to check if the data is stationary.
- **Step 2: Model Identification:** Choose appropriate values for p, d, and q (the parameters of ARIMA). This can be done using ACF (Autocorrelation Function) and PACF (Partial Autocorrelation Function) plots.
- **Step 3: Model Estimation:** Fit the ARIMA model using past data.
- **Step 4: Model Diagnosis:** Check the residuals to ensure the model fits well.
- **Step 5: Forecast:** Use the fitted model to predict future values.

Let $\hat{y}_{n+t}$ denote point forecast made at time origin of the time series value in time period $n+T$ where T greater than or equal to one.

- for example if we have MA(1). The model that uses to forecast the prediction value is:

$$\hat{y}_{n+t}(n) = y_n + \hat{a}_{n+t} - \theta_1 \hat{a}_n$$

- $100(1 - \alpha)\%$ prediction interval calculate

$$\hat{y}_{n+t}(n) \pm t_{\alpha/2}(n-p)SE_{n+t}(n)$$

Where $SE_{n+t}(n)$ is called the standard error of forecast error.

Support Vector Regression (SVR) for Forecasting:

- The Support Vector Regression (SVR) model is a machine learning technique used for prediction and forecasting. It is effective in handling complex and non-linear relationships in data.
- SVR is based on the following key components:
 - **Kernel Function:** Transforms the data into a higher-dimensional space to capture non-linear patterns.
 - **Cost (C):** Controls the trade-off between model complexity and prediction error.
 - **Epsilon (ϵ):** Defines the margin of tolerance where no penalty is given to small errors.

Steps in SVR Forecasting:

- **Step 1: Data Preparation:**
Prepare the dataset by selecting input (independent variables) and output (dependent variable), and divide the data into training and testing sets.
- **Step 2: Parameter Tuning:**
Select the optimal values for parameters such as cost (C) and epsilon (ϵ) using techniques like cross-validation.
- **Step 3: Model Training:**
Train the SVR model using the training data to learn the relationship between variables.
- **Step 4: Model Evaluation:**
Evaluate the model performance using measures such as RMSE and MAE to assess prediction accuracy.
- **Step 5: Prediction:**
Use the trained model to predict future values based on new or unseen data.

SVR-ARIMA Hybrid Model for Forecasting:

- The SVR-ARIMA hybrid model combines statistical and machine learning approaches to improve forecasting accuracy.
- This model uses ARIMA to capture linear patterns in the data and SVR to model non-linear relationships.
- The hybrid model is based on:
 - **ARIMA Component:** Models the linear structure of the time series.
 - **SVR Component:** Captures non-linear patterns from the residuals of the ARIMA model.
 - **Residual Modeling:** The difference between actual and ARIMA predictions is modeled using SVR.

Steps in SVR-ARIMA Forecasting:

- **Step 1: Fit ARIMA Model:**
Apply the ARIMA model to the original time series data to capture the linear components.
- **Step 2: Calculate Residuals:**
Obtain the residuals by subtracting ARIMA predictions from the actual values.
- **Step 3: Train SVR Model:**
Use the residuals as the target variable and train the SVR model to learn non-linear patterns.
- **Step 4: Combine Models:**
Add the ARIMA predictions and SVR predictions of residuals to obtain the final forecast.
- **Step 5: Model Evaluation:**
Evaluate the hybrid model using performance measures such as RMSE and MAE.

Chapter 3

Results and discussion

Multiple Linear Regression

3.1 Multiple Linear Regression Model

Multiple Linear Regression is used to examine the relationship between one dependent variable and two or more independent variables. In this study, oil price is used as the dependent variable, while production and exports are used as independent variables.

The formula for a multiple regression model is:

$$Price = \beta_0 + \beta_1(Production) + \beta_2(Exports) + \varepsilon$$

The parameters of the multiple linear regression model are estimated using the Ordinary Least Squares (OLS) method. This method estimates the values of the coefficients by minimizing the difference between the actual oil prices and the predicted oil prices.

3.2 Estimation of the model Parameter:

The regression coefficients were estimated using the least squares method. The results are shown in the table below:

Variable	Estimate	Std. Error	t value	Pr(> t)
Intercept	-1.213e+02	3.099e+01	-3.915	0.000188
Production	8.038e-03	1.099e-03	7.316	1.65e-10
Exports	-2.331e-03	8.686e-04	-2.683	0.008832

The results show **that** production has a positive and significant effect on oil prices. The coefficient for production is 0.008038, meaning that when production increases by one unit, oil price is expected to increase by 0.008038, while keeping exports constant. Since the p-value is very small (1.65e-10), production is statistically significant.

Exports have a negative and significant effect on oil prices. The coefficient for exports is -0.002331, meaning that when exports increase by one unit, oil price is expected to decrease by 0.002331, while keeping production constant. The p-value is 0.008832, which is less than 0.05, so exports are also statistically significant.

The model has an R-squared value of 0.3987, meaning that about 39.87% of the variation in oil prices is explained by production and exports. The overall model is significant because the F-test p-value is 1.127e-09, which is less than 0.05.

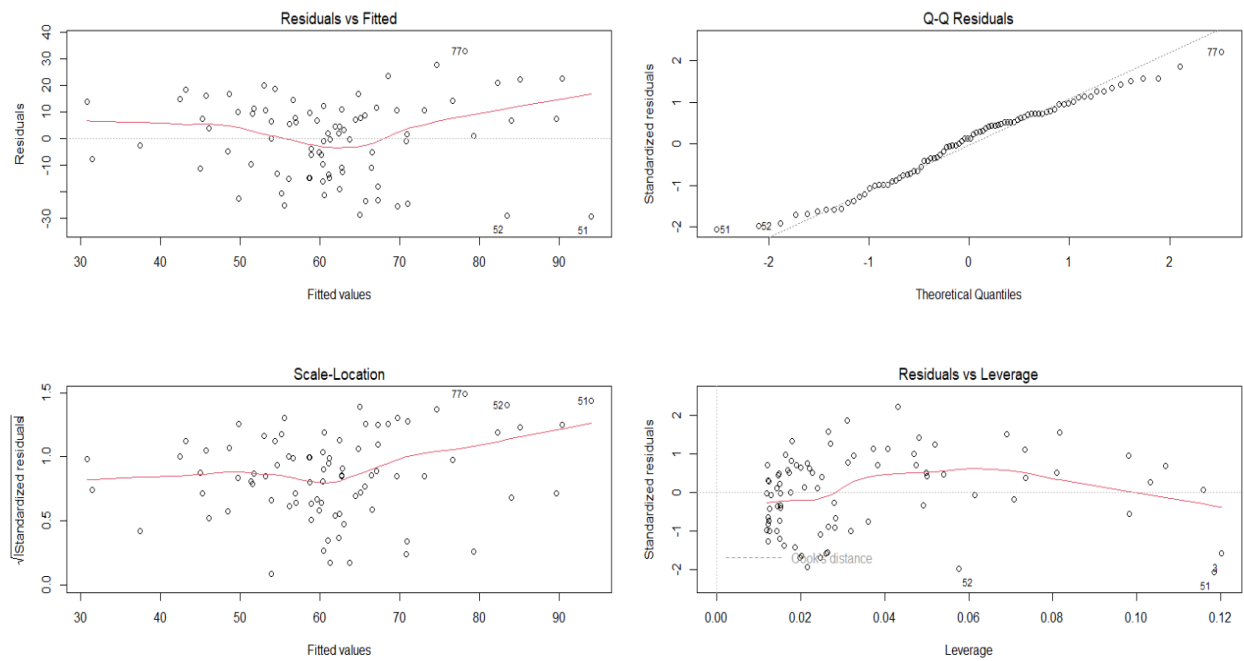


Figure 3.1: four plots are diagnostic checks for the assumptions of linear regression.

1. Residuals vs Fitted

This plot is used to check linearity and model fit.

- The points are randomly scattered around zero.
- There is no clear pattern or curve.

This indicates that the linear model is appropriate, and the relationship between variables is reasonably linear. However, there is slight dispersion at higher fitted values, suggesting minor variation.

2. Q-Q Residuals

This plot checks the normality of residuals.

- Most points lie close to the straight line.
- A few points deviate slightly at the ends.

The residuals are approximately normally distributed, although there are small deviations in the tails. Overall, the normality assumption is reasonably satisfied.

3. Scale-Location Plot

This plot is used to check homoscedasticity (constant variance).

- The points are fairly spread, but there is a slight upward trend.

There is some indication of non-constant variance (heteroscedasticity), as the spread of residuals increases slightly with fitted values. This suggests that the assumption of equal variance may be mildly violated.

4. Residuals vs Leverage

This plot identifies influential observations.

- Most points are within the safe region.
- No points strongly exceed Cook's distance lines.

There are no highly influential observations that strongly affect the model. The model is stable and not overly influenced by extreme values.

The regression model generally satisfies the main assumptions:

- Linearity
- Normality (approximately)
- No strong influential points
- However, there is slight heteroscedasticity, which means the variance is not perfectly constant.

For fix heteroscedasticity I apply log transformation

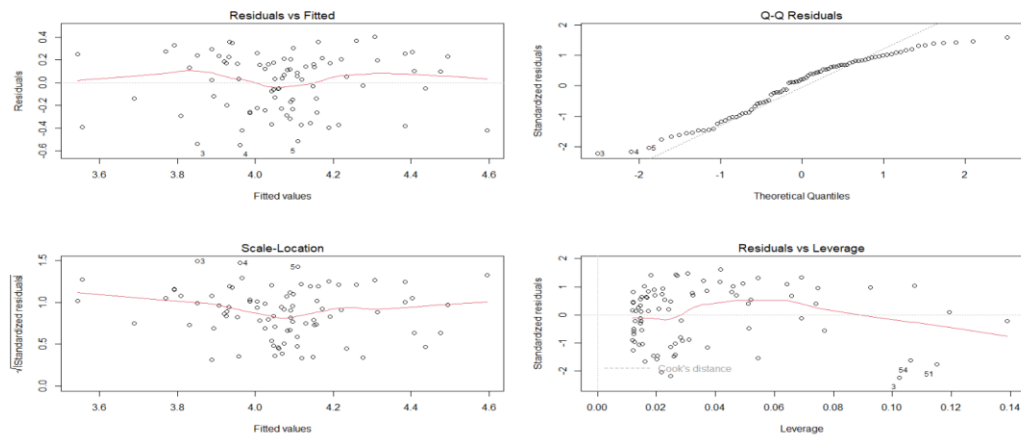
The log regression model shows that both variables are statistically significant:

Variable	Estimate	p-value	Interpretation
log_productions	3.8320	1.82e-09	Positive significant effect
log_exports	-1.2010	0.00192	Negative significant effect

The model is significant overall because the F-test p-value = $1.229\text{e-}08$, which is less than 0.05.

The R-squared = 0.3622, meaning the model explains about 36.22% of the variation in oil prices.

Diagnostic plots



The plots look acceptable:

- Residuals vs Fitted: residuals are mostly scattered randomly, so linearity is acceptable.
- Q-Q Plot: residuals are close to the line, so normality is mostly acceptable.
- Scale-Location: the spread is more stable than before, so heteroscedasticity is reduced.
- Residuals vs Leverage: no serious influential points appear.

After applying the log transformation, the regression assumptions improved, especially the constant variance assumption. The model remains statistically significant, and both production and exports have significant effects on oil prices. Therefore, the log-transformed model can be considered more suitable than the original model.

3.3.1 Hypothesis Testing for 2016 Data in Multiple Linear Regression

For 2016 data, multiple linear regression is applied to test whether production and exports have a significant effect on oil prices. The model is:

$$Price = \beta_0 + \beta_1(Production) + \beta_2(Exports) + \varepsilon$$

1. t-test

The t-test is used to test the individual effect of each predictor, which are production and exports, on oil prices.

Hypotheses:

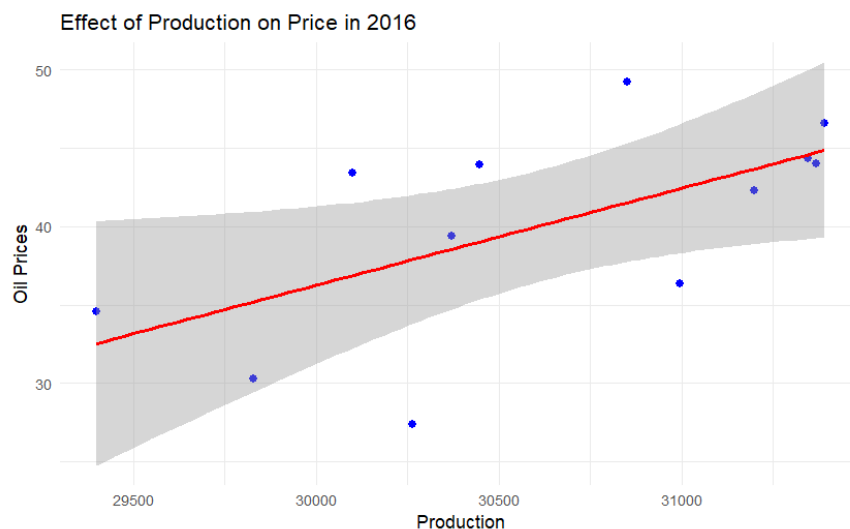
H₀ (Null Hypothesis): The (Production or Exports) has no effect on Price ($\beta = 0$).

H₁ (Alternative Hypothesis): The (Production or Exports) significantly affects Price ($\beta \neq 0$).

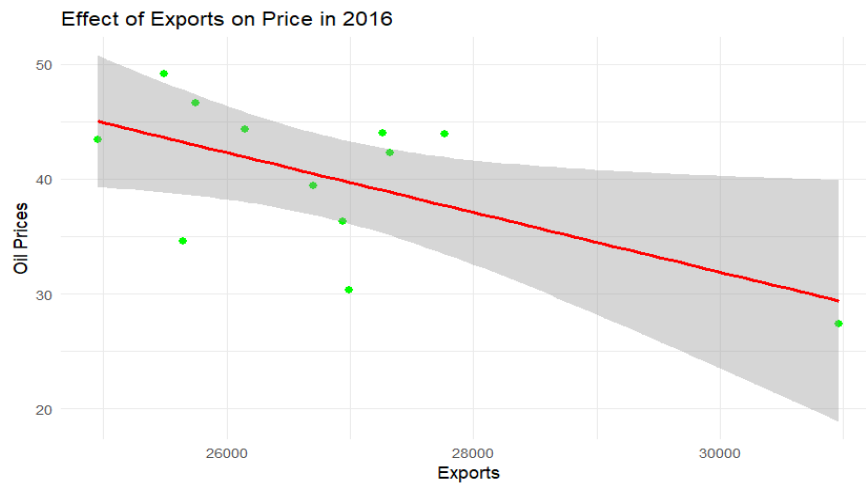
t-test Results

Variable	Estimate	Std. Error	t value	p-value
Intercept	-7.633e+01	5.795e+01	-1.317	0.22037
Production	6.021e-03	1.759e-03	3.423	0.00759
Exports	-2.532e-03	7.448e-04	-3.400	0.00787

For production, the p-value is 0.00759, which is less than 0.05. Therefore, the null hypothesis is rejected. This means production has a significant positive effect on oil prices in 2016.



For exports, the p-value is 0.00787, which is also less than 0.05. Therefore, the null hypothesis is rejected. This means exports have a significant effect on oil prices in 2016. Since the coefficient is negative, exports have a negative relationship with oil prices.



2. F-test

The F-test is used to test the overall significance of the regression model.

Hypotheses:

H_0 : (Null Hypothesis): The model has no explanatory power (all coefficients = 0).

H_1 : (Alternative Hypothesis): At least one predictor significantly affects Price (at least one coefficient $\neq 0$).

F-test Results

Statistic	Value
F-statistic	11.95
p-value	0.002927
R-squared	0.7265
Adjusted R-squared	0.6657

The F-test p-value is 0.002927, which is less than 0.05. Therefore, the overall regression model is statistically significant. This means that production and exports together explain a significant part of the variation in oil prices in 2016.

The R-squared value is 0.7265, meaning the model explains about 72.65% of the variation in oil prices for 2016. Overall, the results show that both production and exports significantly affect oil prices in 2016.

3.3.2 Hypothesis Testing for 2017–2018 Data in Multiple Linear Regression

1- t-test

t-test for individual predictors

For each predictor (Production & Exports), the hypotheses are:

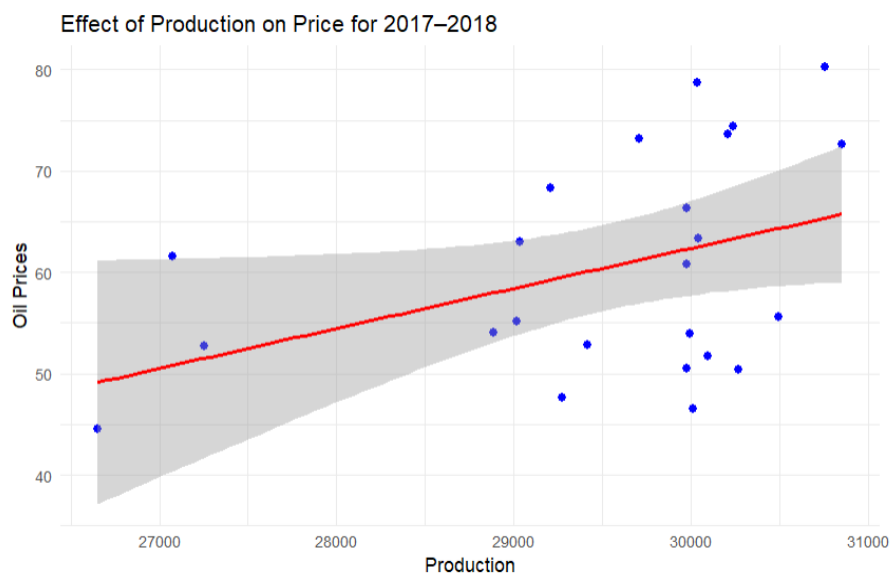
H_0 (Null Hypothesis): The (Production or Exports) has no effect on Price ($\beta = 0$).

H_1 (Alternative Hypothesis): The (Production or Exports) significantly affects Price ($\beta \neq 0$).

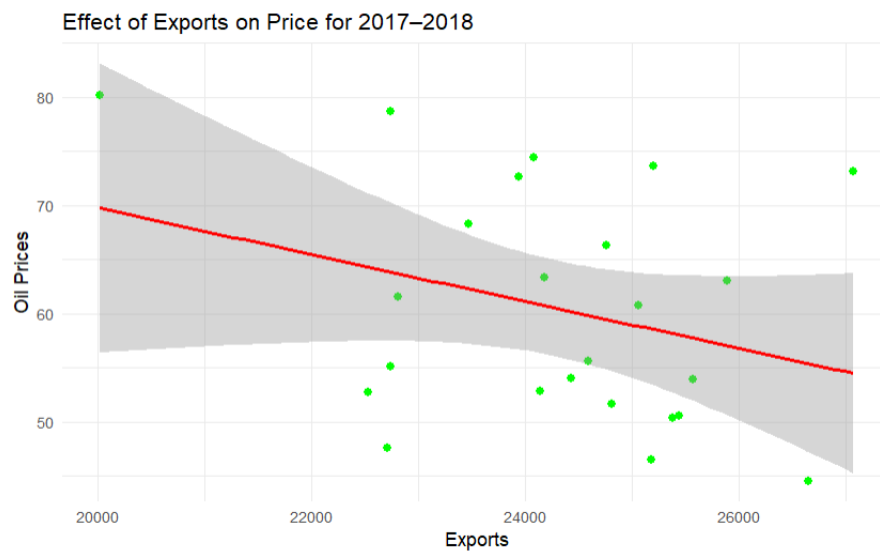
Results Table

Variable	Estimate	Std. Error	t value	Pr(> t)
Intercept	-3.227	63.029	-0.051	0.9597
Production	0.003903	0.001823	2.141	0.0442
Exports	-0.002118	0.001312	-1.615	0.1213

For Production, the p-value is 0.0442, which is less than 0.05. This means we reject the null hypothesis. Therefore, production has a statistically significant positive effect on oil prices for 2017–2018.



For Exports, the p-value is 0.1213, which is greater than 0.05. This means we fail to reject the null hypothesis. Therefore, exports do not have a statistically significant effect on oil prices for 2017–2018.



2- F-test

H_0 : (Null Hypothesis): The model has no explanatory power (all coefficients = 0).

H_1 : (Alternative Hypothesis): At least one predictor significantly affects Price (at least one coefficient $\neq 0$).

F-test Results

Statistic	Value
F-statistic	3.652
p-value	0.04353
R-squared	0.2581
Adjusted R-squared	0.1874

The F-test p-value is 0.04353, which is less than 0.05. Therefore, the overall regression model is statistically significant.

This means that the independent variables (production and exports) together explain a significant portion of the variation in oil prices for 2017–2018.

However, the R-squared value is 0.2581, which indicates that the model explains about 25.81% of the variation in oil prices. This is lower than the 2016 model, suggesting weaker explanatory power. Production has a significant effect on oil prices, while exports do not show a significant effect for 2017–2018

3.3.3 Hypothesis Testing for 2019–2021 Data in Multiple Linear Regression

1- t-test

t-test for individual predictors

For each predictor (Production & Exports), the hypotheses are:

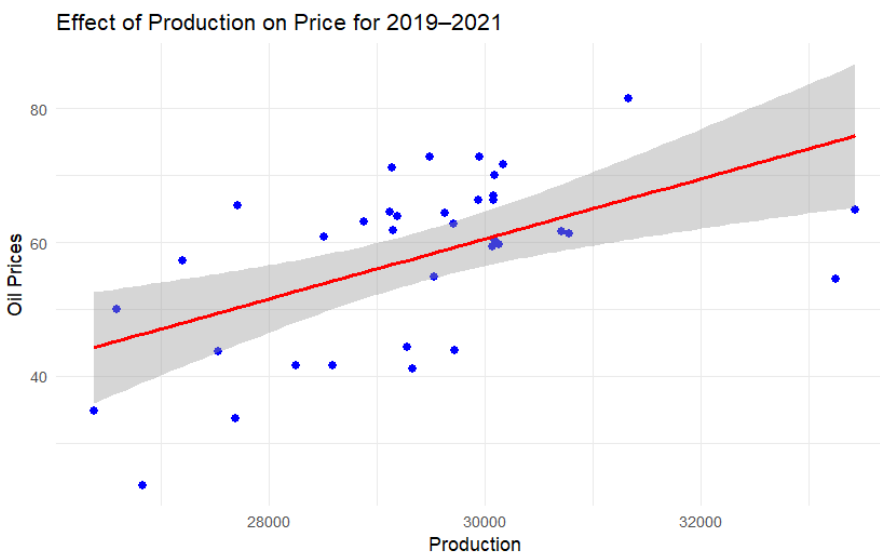
H_0 (Null Hypothesis): The (Production or Exports) has no effect on Price ($\beta = 0$).

H_1 (Alternative Hypothesis): The (Production or Exports) significantly affects Price ($\beta \neq 0$).

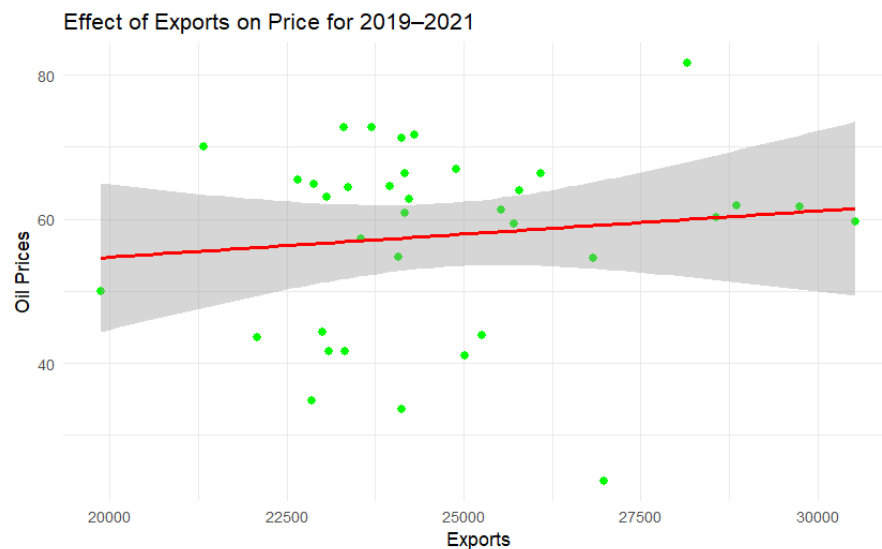
Results Table

Variable	Estimate	Std. Error	t value	Pr(> t)
Intercept	-6.953e+01	3.648e+01	-1.906	0.06536
Production	4.834e-03	1.320e-03	3.661	0.000871
Exports	-5.955e-04	8.776e-04	-0.679	0.50215

For Production, the p-value is 0.000871, which is less than 0.05. Therefore, the null hypothesis is rejected. This means production has a statistically significant positive effect on oil prices for 2019–2021.



For Exports, the p-value is 0.50215, which is greater than 0.05. Therefore, we fail to reject the null hypothesis. This means exports do not have a statistically significant effect on oil prices for 2019–2021



2- F-test

H_0 : (Null Hypothesis): The model has no explanatory power (all coefficients = 0).

H_1 : (Alternative Hypothesis): At least one predictor significantly affects Price (at least one coefficient $\neq 0$).

F-test Results

Statistic	Value
F-statistic	7.015
p-value	0.002894
R-squared	0.2983
Adjusted R-squared	0.2558

The F-test p-value is 0.002894, which is less than 0.05. Therefore, the overall regression model is statistically significant.

This means that the independent variables (production and exports) together explain a significant portion of the variation in oil prices for 2019–2021.

The R-squared value is 0.2983, meaning the model explains about 29.83% of the variation in oil prices. This indicates moderate explanatory power. Production has a significant effect on oil prices, while exports do not show a significant effect for 2019–2021.

3.3.4 Hypothesis Testing for 2022 Data in Multiple Linear Regression

1- t-test

t-test for individual predictors

For each predictor (Production & Exports), the hypotheses are:

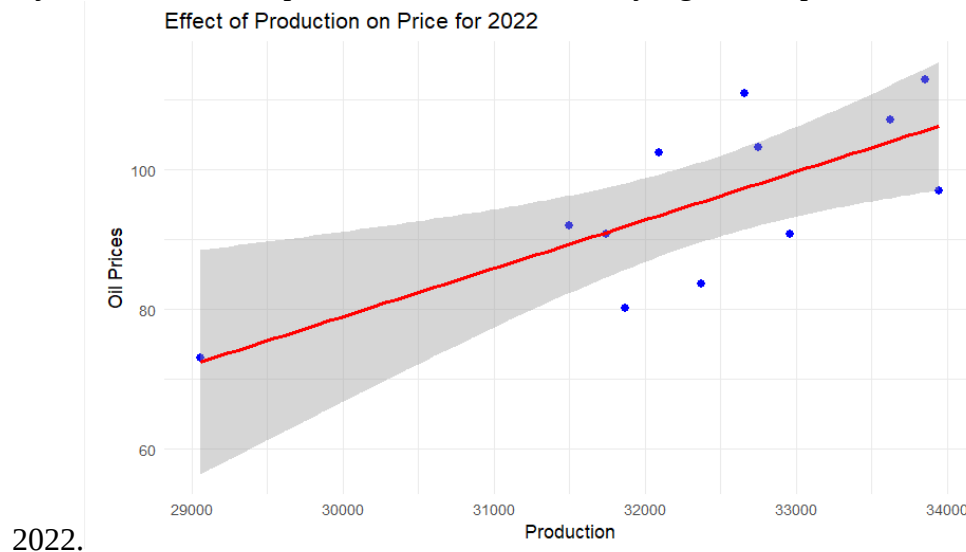
H_0 (Null Hypothesis): The (Production or Exports) has no effect on Price ($\beta = 0$).

H_1 (Alternative Hypothesis): The (Production or Exports) significantly affects Price ($\beta \neq 0$).

Results Table

Variable	Estimate	Std. Error	t value	Pr(> t)
Intercept	-84.707	85.572	-0.990	0.34809
Production	0.007242	0.002106	3.438	0.00741
Exports	-0.002051	0.002497	-0.821	0.43266

For Production, the p-value is 0.00741, which is less than 0.05. Therefore, the null hypothesis is rejected. This means production has a statistically significant positive effect on oil prices for



For Exports, the p-value is 0.43266, which is greater than 0.05. Therefore, we fail to reject the null hypothesis. This means exports do not have a statistically significant effect on oil prices for 2022.



2- F-test

H_0 : (Null Hypothesis): The model has no explanatory power (all coefficients = 0).

H_1 : (Alternative Hypothesis): At least one predictor significantly affects Price (at least one coefficient $\neq 0$).

F-test Results

Statistic	Value
F-statistic	5.925
p-value	0.02281
R-squared	0.5684
Adjusted R-squared	0.4724

The F-test p-value is 0.02281, which is less than 0.05. Therefore, the overall regression model is statistically significant.

This means that production and exports together explain a significant portion of the variation in oil prices for 2022. The R-squared value is 0.5684, meaning the model explains about 56.84% of the variation in oil prices, which indicates a relatively stronger model compared to previous periods. Production has a significant effect on oil prices, while exports do not show a significant effect for 2022.

3.4 Confidence Intervals in Multiple Regression

3.4.1- 95% Confidence Intervals for Full Data

Predictor	2.5%	97.5%
Intercept	-182.96	-59.65
Production	0.00585	0.01022
Exports	-0.00406	-0.00060

The confidence interval for production does not include zero, so production has a significant positive effect on oil prices. The confidence interval for exports also does not include zero, so exports have a significant negative effect on oil prices.

3.4.2- 95% Confidence Intervals for 2016

Predictor	2.5%	97.5%
Intercept	-207.42	54.77
Production	0.00204	0.00999
Exports	-0.00422	-0.00085

For 2016, the confidence interval for production does not include zero, meaning production significantly affects oil prices positively. The interval for exports also does not include zero, meaning exports significantly affect oil prices negatively.

3.4.3- 95% Confidence Intervals for 2017–2018

Predictor	2.5%	97.5%
Intercept	-134.30	127.85
Production	-0.00117	0.00769
Exports	-0.00485	0.00061

For 2017–2018, the confidence intervals for production and exports include zero. This means their effects are not clearly significant at the 95% confidence level.

3.4.4 95% Confidence Intervals for 2019–2021

Predictor	2.5%	97.5%
Intercept	-143.74	4.68
Production	0.00215	0.00752
Exports	-0.00238	0.00119

For 2019–2021, the confidence interval for production does not include zero, so production has a significant positive effect on oil prices. However, the interval for exports includes zero, meaning exports do not show a significant effect.

3.4.5 95% Confidence Intervals for 2022

Predictor	2.5%	97.5%
Intercept	-278.24	108.69
Production	0.00248	0.01201
Exports	-0.00770	0.00360

For 2022, the confidence interval for production does not include zero, indicating a significant positive effect on oil prices. The confidence interval for exports includes zero, so exports are not statistically significant in this period.

Chapter 4

Results and Discussion

Time Series

Introduction

Time series analysis is an important statistical technique used to analyze data collected over time. In this study, time series methods are applied to examine the behavior of oil prices and to identify patterns such as trends and fluctuations over the observed period.

The main objective of this analysis is to understand how oil prices change over time and to build an appropriate model that can be used for forecasting future values. This is achieved by first visualizing the data, then testing for stationarity, and finally applying suitable time series models based on the Box–Jenkins methodology.

By analyzing the time-dependent structure of the data, more accurate predictions can be made, which are useful for economic planning and decision-making.

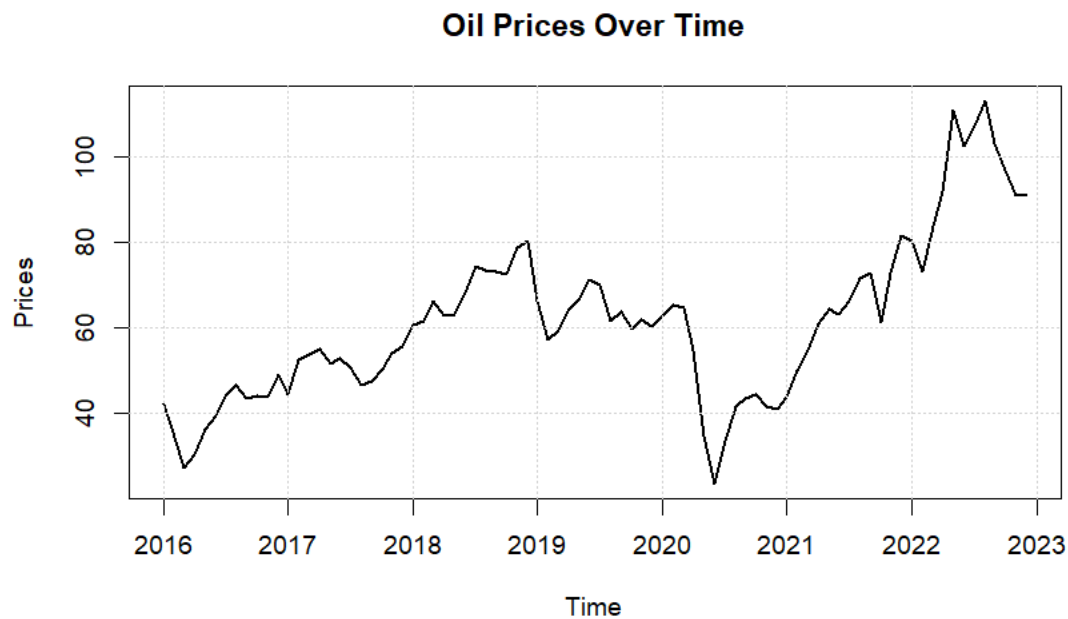


Figure 4.1: Trend of Oil Prices in Oman (2016–2022)

The figure illustrates the movement of oil prices over time from 2016 to 2022. The plot shows an overall upward trend, although it is characterized by noticeable fluctuations throughout the period. In the early years (2016–2018), oil prices gradually increase with some minor variations. Around 2019, the prices show moderate instability, followed by a sharp decline in 2020, which represents a significant drop in oil prices.

After 2020, oil prices begin to recover steadily, showing a strong upward trend during 2021 and reaching peak levels in 2022. This sharp increase in the most recent years indicates the influence of external factors such as global economic recovery and market conditions.

The time series demonstrates periods of growth, decline, and recovery, highlighting the volatile nature of oil prices. The absence of a clear seasonal pattern suggests that non-seasonal time series models may be more appropriate for analyzing this data. This analysis provides a better understanding of oil price behavior over time and supports future forecasting and decision-making related to the oil sector.

4.1 Using the SAC to Find a Stationary Time Series

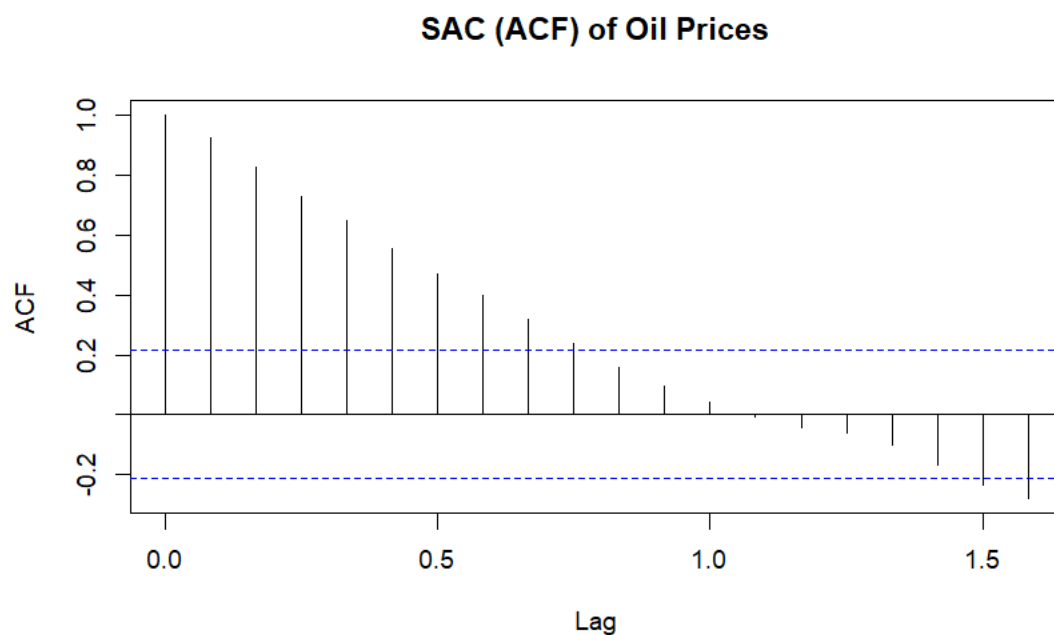


Figure 4.2: Autocorrelation Function (ACF) of Oil Prices

The SAC (ACF) plot of oil prices shows that the autocorrelation values start very high and decrease slowly over many lags. This gradual decay indicates that the observations are strongly correlated with their past values for a long period.

This pattern suggests that the time series is non-stationary, because in a stationary series the autocorrelations usually drop quickly to zero. The presence of a slow decline in the ACF indicates a trend in the data and confirms that the mean is not constant over time.

Therefore, the original oil price series does not satisfy the stationarity condition required for time series modeling. To address this issue, differencing should be applied to remove the trend and make the series stationary.

Apply differencing:

$$Z_t = Y_t - Y_{t-1}$$

where Z_t represents the differenced version of the original series, making it stationary.

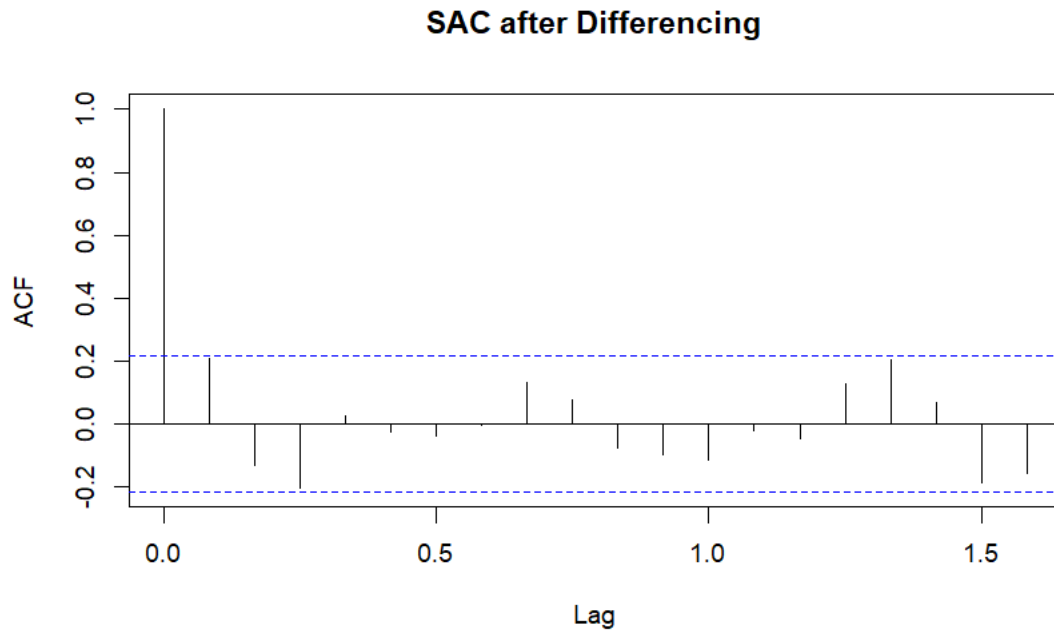


Figure 4.3: ACF Plot of First-Differenced Oil Prices

The SAC (ACF) plot after first differencing shows that most autocorrelation values drop quickly and lie within the confidence bounds, with only a few small spikes. Unlike the original series, there is no slow decay pattern.

This indicates that the strong dependence between observations has been removed, and the series now fluctuates randomly around a constant mean. Therefore, the differenced series can be considered stationary.

The result confirms that applying first differencing ($d = 1$) was sufficient to achieve stationarity. This suggests that an ARIMA model with $d = 1$ is appropriate for modeling the oil price series.

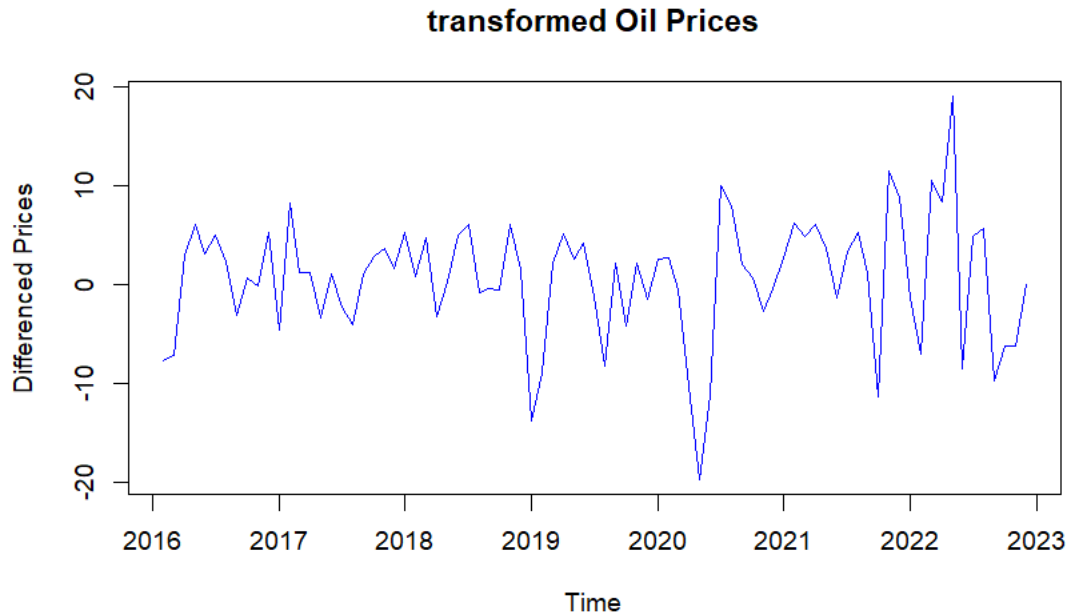


Figure 4.4: Transformed (Stationary) Prices of the Series

The transformed time series plot shows the oil prices after applying first differencing. The series fluctuates around zero without a clear upward or downward trend, indicating that the long-term trend has been successfully removed.

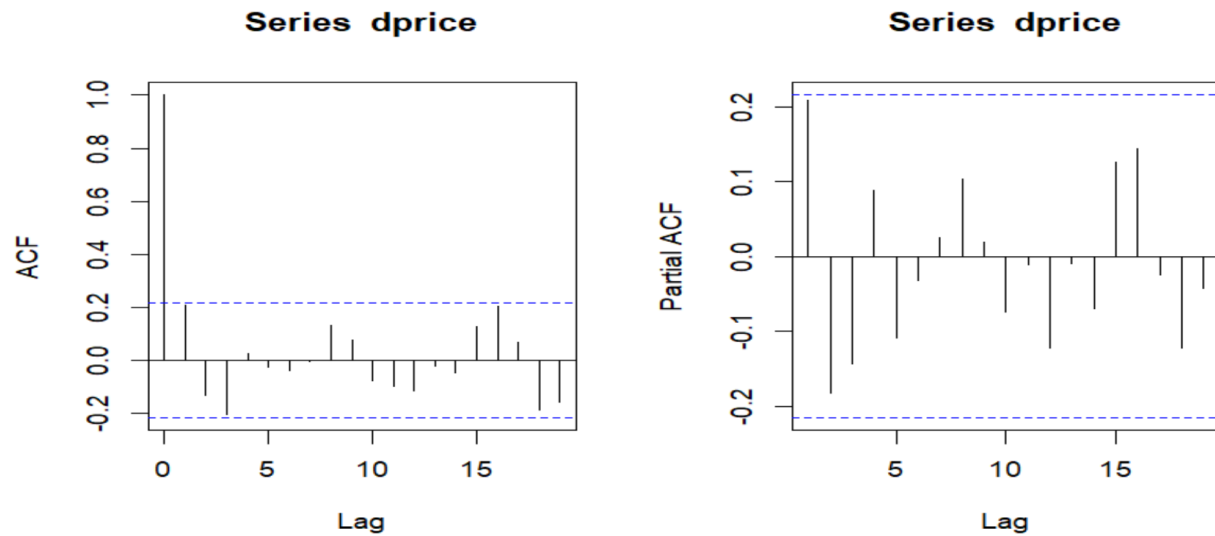
The values vary randomly over time with relatively stable variance, although some sharp spikes are still observed, particularly around 2020 and 2022. These spikes represent sudden changes in oil prices, likely due to external economic or global events.

ACF

- Most values are small and within the blue lines
- No slow decay

This confirms:

The series is now stationary



The ACF and PACF plots of the differenced series show that most autocorrelation values lie within the confidence bounds, indicating that the series has become stationary after first differencing. The ACF shows small spikes, while the PACF does not display any significant pattern.

PACF

- No clear big spikes
- Values are mostly small

This means:

- No strong AR pattern
- Model is likely simple

4.2 ARIMA Model (Box–Jenkins Methodology)

4.2.1- Model Identification

The first step in the Box–Jenkins methodology is to identify a suitable model for the time series. This process involves examining the behavior of the series and analyzing the autocorrelation structure using the Sample Autocorrelation Function (SAC/ACF) and the Partial Autocorrelation Function (PACF).

The initial analysis of the oil price series showed a clear trend, indicating that the data was non-stationary. The SAC plot of the original series displayed a slow decay in autocorrelation values, which confirms the presence of non-stationarity.

To address this issue, first differencing was applied to the data. After differencing, the SAC plot showed that the autocorrelations dropped quickly and most values fell within the confidence bounds. This indicates that the differenced series is stationary.

Further analysis using ACF and PACF plots of the differenced series showed that:

- The ACF contains small spikes with no clear pattern.
- The PACF does not show any significant spikes.

Based on these observations, the time series does not exhibit a strong autoregressive pattern, and only a simple moving average component may be present. Therefore, an ARIMA(0,1,1) model is suggested as an appropriate model for the oil price series.

This model will be estimated and evaluated in the next section.

4.2.2- Model Estimation

The ARIMA(0,1,1) model was estimated using the differenced oil price series. The results are presented below:

Model Results

Parameter	Estimate	Std. Error
MA(1)	0.2506	0.1016

The estimated coefficient for the moving average term MA(1) is 0.2506, which indicates a positive relationship between the current value and past error terms. This suggests that previous shocks have a moderate influence on current oil prices.

The variance of the residuals (σ^2) is 36.47, indicating the level of unexplained variation in the model. The log-likelihood value is -266.55, which is used to evaluate model fit.

The information criteria values are:

- AIC = 537.11
- AICc = 537.26
- BIC = 541.94

Model Accuracy

Measure	Value
RMSE	5.966
MAE	4.547
MAPE	8.081
ACF1	-0.0096

The error measures indicate that the model has a reasonable level of accuracy. The MAPE value (8.08%) suggests that the forecasting error is relatively low. Additionally, the ACF of residuals is close to zero, indicating that there is no significant autocorrelation remaining in the residuals.

The ARIMA(0,1,1) model provides a good fit for the oil price data. The model captures the main patterns in the series, and the residual diagnostics suggest that the assumptions are reasonably satisfied. Therefore, this model can be used for forecasting future oil prices.

Box-Pierce Test ,model diagnostic checking

The Box-Pierce test is used to check whether the residuals of the model are independent (no autocorrelation).

Hypotheses

H₀: Residuals are independent (no autocorrelation)

H₁: Residuals are autocorrelated

Result : p-value = 0.783

Since the p-value (0.783) is greater than 0.05, we fail to reject the null hypothesis. This indicates that there is no significant autocorrelation in the residuals. This means that the residuals behave like white noise, and the model has successfully captured the underlying pattern in the data.

4.2.3- Forecasting

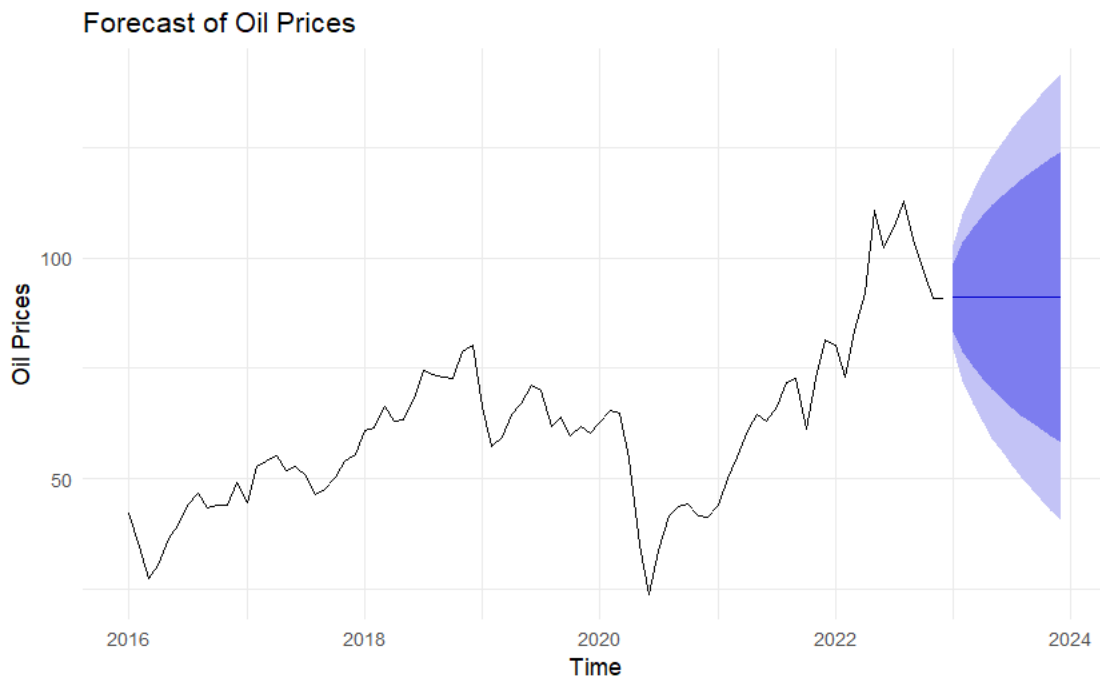


Figure: Forecast of Oil Prices

The forecast plot presents the predicted values of oil prices for the next 12 periods based on the $ARIMA(0,1,1)$ model. The central line represents the forecasted oil prices, while the shaded regions indicate the confidence intervals.

The forecast shows that oil prices are expected to remain relatively stable around the current level, with no strong upward or downward trend. This suggests that the sharp fluctuations observed in previous years may not continue in the short term.

The shaded areas (confidence intervals) widen as time progresses, indicating increasing uncertainty in future predictions. The darker region represents a narrower confidence interval, while the lighter region represents a wider interval. This means that short-term forecasts are more reliable than long-term forecasts.

the model suggests that oil prices will stabilize in the near future, although variability is still expected. The results provide useful insights for understanding possible future movements in oil prices, but they should be interpreted carefully due to the influence of external economic and global factors.

	Point Forecast <dbl>	Lo 80 <dbl>	Hi 80 <dbl>	Lo 95 <dbl>	Hi 95 <dbl>
Jan 2023	91.1211	83.38203	98.86016	79.28522	102.9570
Feb 2023	91.1211	78.72894	103.51325	72.16893	110.0733
Mar 2023	91.1211	75.39729	106.84490	67.07361	115.1686
Apr 2023	91.1211	72.65734	109.58486	62.88321	119.3590
May 2023	91.1211	70.27444	111.96775	59.23889	123.0033
Jun 2023	91.1211	68.13729	114.10490	55.97040	126.2718
Jul 2023	91.1211	66.18262	116.05958	52.98098	129.2612
Aug 2023	91.1211	64.37039	117.87180	50.20942	132.0328
Sep 2023	91.1211	62.67338	119.56882	47.61406	134.6281
Oct 2023	91.1211	61.07205	121.17014	45.16505	137.0771

The forecast results show that the predicted oil prices remain constant at approximately 91.12 USD over the forecast period. This indicates that the ARIMA(0,1,1) model expects oil prices to stabilize in the short term.

The forecast intervals provide a range of possible values. The 80% confidence interval is narrower, while the 95% confidence interval is wider, indicating greater uncertainty. As the forecast horizon increases, the intervals become wider, which reflects increasing uncertainty in future predictions.

For example, in January 2023, the oil price is expected to lie between 79.29 and 102.96 (95% interval). By October 2023, this range expands significantly from 45.17 to 137.08, showing higher uncertainty over time.

4.2.4- ARIMA(1,1,0)

The data is divided into:

- Training data (75%) → used to build the model
- Testing data (25%) → not used here yet

Model Results

Item	Value
Model	ARIMA(1,1,0)
AR(1) coefficient	0.3485
Std. Error	0.1205
AIC	382.81
RMSE	5.088
MAE	3.847
MAPE	8.017%
ACF1	0.0948

The ARIMA(1,1,0) model was estimated using the training portion of the oil price data. The AR(1) coefficient is 0.3485, which indicates a positive relationship between the current differenced value and the previous one. This means that past changes in oil prices have a moderate influence on current changes.

The model accuracy measures show that the RMSE is 5.088 and the MAE is 3.847, while the MAPE is 8.017%, indicating that the model provides a reasonable level of forecasting accuracy. The ACF1 value is 0.0948, which is close to zero, suggesting that there is no significant autocorrelation remaining in the residuals.

4.3 -Support Vector Regression (SVR)

Model Estimation

The Support Vector Regression (SVR) model was applied to the oil price data using a training dataset. The model parameters were selected using a tuning process to find the optimal values.

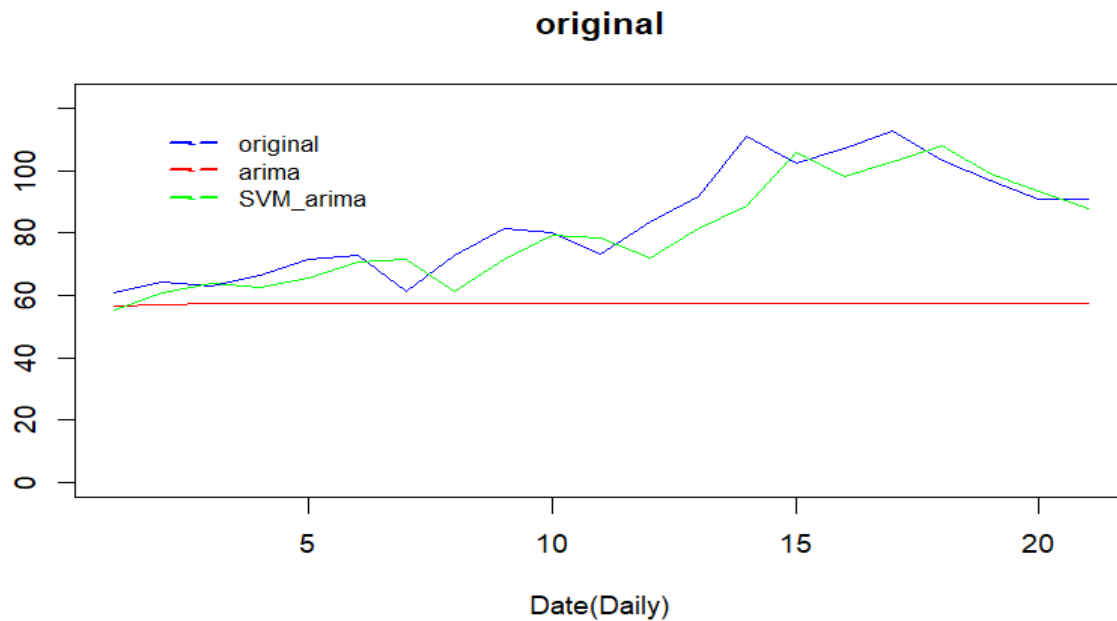
SVR Results

Parameter	Value
Cost	50
Epsilon	0.25
Accuracy Measure	Value
RMSE	37.02
MAE	27.11

The SVR model selected cost = 50 and epsilon = 0.25 as the best parameters. The cost value shows that the model gives stronger importance to reducing prediction errors, while epsilon allows small errors within a tolerance range.

The accuracy results show that the SVR model has relatively high errors, with RMSE = 37.02 and MAE = 27.11. This means the predicted oil prices are not very close to the actual values.

Overall, the SVR model does not perform well for this dataset compared with the ARIMA model, which had lower error values. Therefore, ARIMA is more suitable for forecasting oil prices in this study.



The RMSE results compare four forecasting methods:

Model	RMSE
ARIMA	31.26
ARIMA Regression	32.14
SVR-ARIMA Linear	8.276
SVR-ARIMA Polynomial	8.276

The results show that the SVR-ARIMA models have much lower RMSE values compared with the ARIMA models. The RMSE for SVR-ARIMA is around 8.28, while ARIMA has an RMSE above 31. This indicates that the SVR-ARIMA model provides more accurate predictions.

The plot compares the original oil prices with ARIMA and SVR-ARIMA forecasts. The blue line represents the original data, the red line represents the ARIMA forecast, and the green line represents the SVR-ARIMA forecast. The ARIMA forecast appears almost flat and does not follow the movement of the original data well. In contrast, the SVR-ARIMA forecast follows the pattern of the original series more closely, especially during periods of increase and decrease.

Conclusion

Based on the RMSE values and the plot, the SVR-ARIMA model performs better than the ARIMA model. It gives lower prediction errors and follows the actual oil price movement more closely. Therefore, the SVR-ARIMA model is more suitable for forecasting oil prices in this study.

1. SVR (alone)

- RMSE ≈ 37
 - Poor performance
- Conclusion: SVR alone is NOT good

2. SVR-ARIMA (hybrid model)

- RMSE ≈ 8.28
- Much better than ARIMA

3. ARIMA

- RMSE ≈ 31
- Worse than SVR-ARIMA

The SVR model alone does not perform well, as it produces high prediction errors. However, when combined with ARIMA in a hybrid model (SVR-ARIMA), the performance improves significantly.

Based on the RMSE results, the SVR-ARIMA model has the lowest error compared to both ARIMA and SVR models. Additionally, the plot shows that the SVR-ARIMA model follows the actual oil price movements more closely than the ARIMA model.

Therefore, the SVR-ARIMA model is the most accurate and suitable model for forecasting oil prices in this study.

Chapter 5

CONCLUSIONS

5.1 Conclusion

This study aimed to analyze and model oil prices in Oman using different statistical and machine learning techniques. The analysis included descriptive statistics, regression models, time series analysis, and forecasting methods.

The results from the multiple linear regression showed that production has a significant positive effect on oil prices, while exports did not show consistent significance across all periods. This indicates that production plays a more important role in influencing oil prices.

In the time series analysis, the oil price data was found to be non-stationary, but it became stationary after applying first differencing. Based on the ACF and PACF analysis, the ARIMA(0,1,1) model was identified and estimated. The model showed acceptable performance and was able to capture the general behavior of oil prices.

However, when comparing forecasting methods, it was found that the SVR model alone did not perform well, as it produced high prediction errors. In contrast, the hybrid SVR-ARIMA model provided the best performance, with the lowest RMSE values and better tracking of actual oil price movements.

Overall, the results show that combining statistical models with machine learning methods improves forecasting accuracy. The SVR-ARIMA model proved to be the most suitable approach for predicting oil prices in this study.

5.2 Recommendations

Based on the findings of this study, the following recommendations are suggested:

- It is recommended to use hybrid models such as SVR-ARIMA for oil price forecasting, as they provide more accurate results compared to single models.
- Policymakers and decision-makers in Oman should consider using advanced forecasting models to better understand oil price movements and support economic planning.
- Future studies can include additional variables such as global demand, geopolitical factors, and economic indicators to improve the accuracy of the models.
- It is also recommended to test other machine learning models such as neural networks or deep learning techniques for further improvement in forecasting performance.
- Continuous monitoring and updating of models is important, as oil prices are affected by rapidly changing global conditions.

This study demonstrates that combining traditional statistical methods with modern machine learning techniques can significantly enhance the accuracy of oil price forecasting and provide valuable insights for economic decision-making.

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